



PicoMagnetometer Setting up a Raspberry Pi 4/5

Pico^{nT}

<https://ukraa.com/>

The UK Radio Astronomy Association

A charitable incorporated organisation

Registered charity in England and Wales No. 1123866



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Set-up instructions

These instructions cover setting up your Raspberry Pi4 or Raspberry Pi 5 with a suitable operating system and ability to be remotely access via VNC.

It is recommended that you use a Raspberry Pi 4/5 with at least 2GB memory.

When using the Raspberry Pi Imager, remember to write down the **hostname** you use – this is used to view the web page with the results from your UKRAA PicoMagnetometer unit.

Please keep the username as **pi** when using OS customisation within the Raspberry Pi Imager.

The instructions for accessing your Raspberry Pi heedlessly assume use of Windows PC and that the following two applications are preinstalled on that PC.

- PuTTY
- RealVNC Viewer

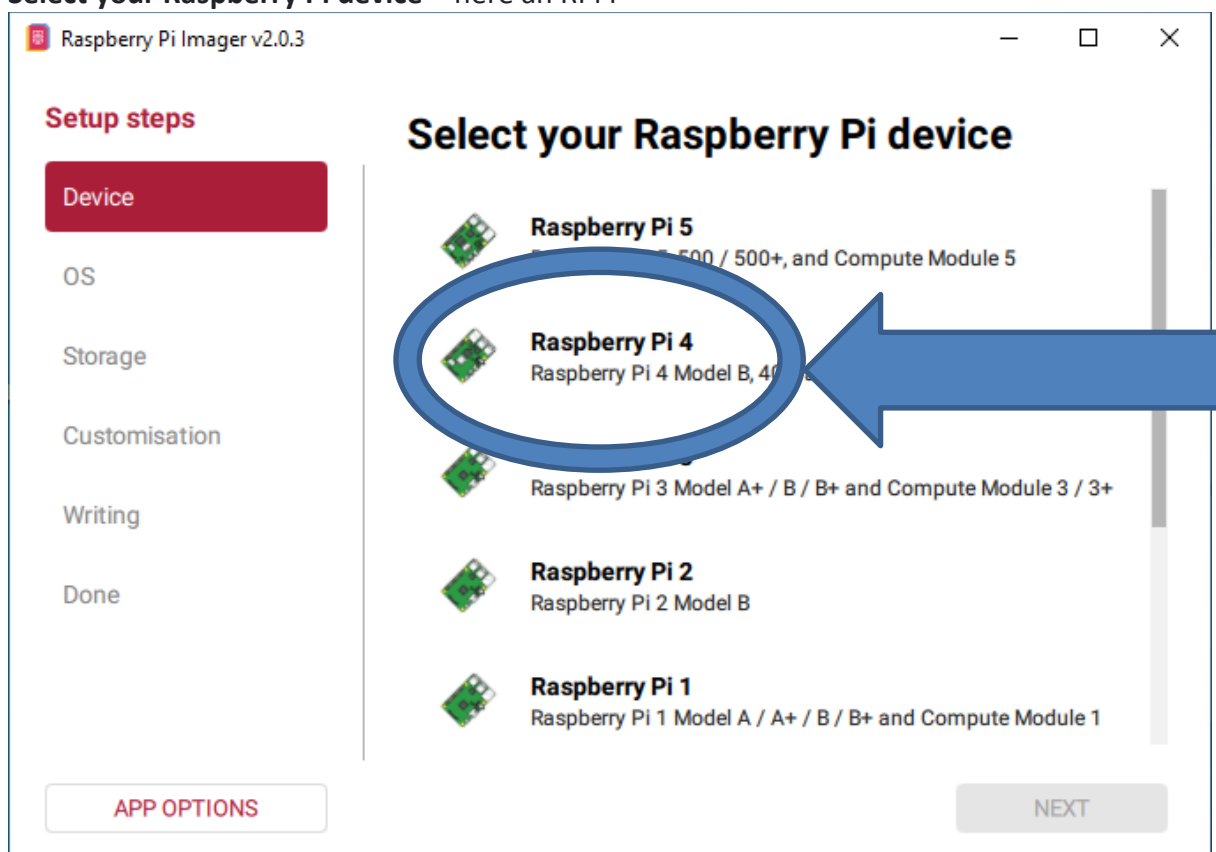
No information on installing these two applications is given in this text – there are numerous guides available on the internet.

Raspberry Pi OS

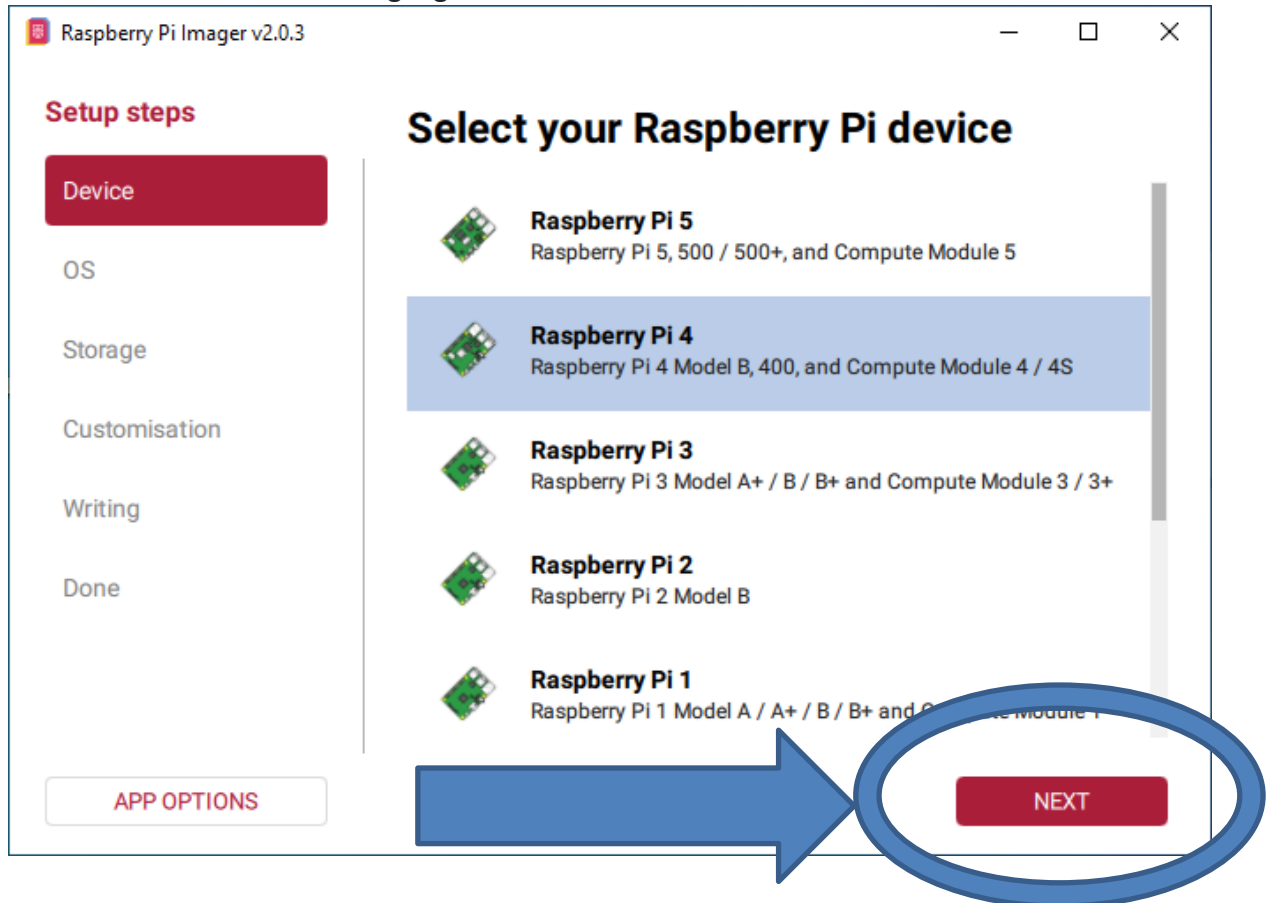
1. Download [Raspberry Pi Imager](#) for your operating system.



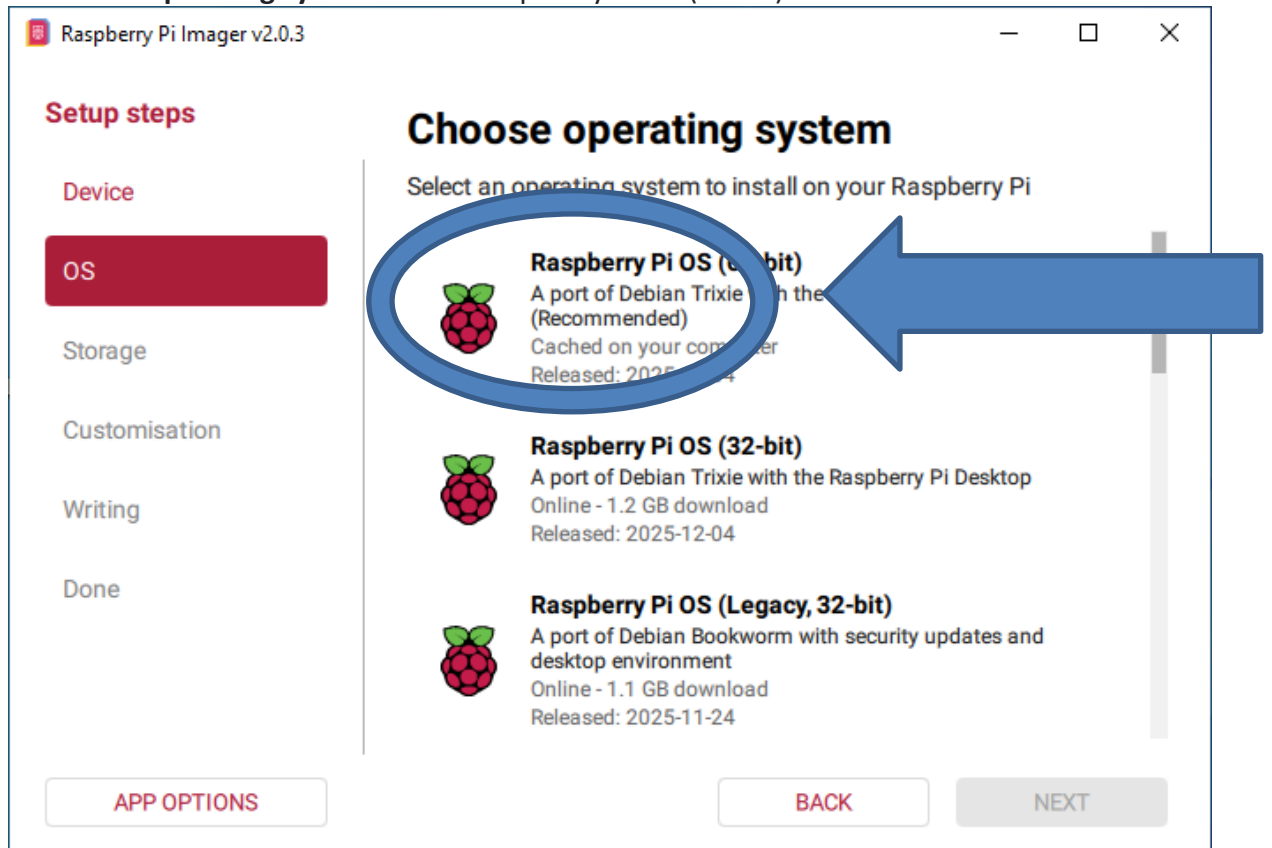
2. Run Raspberry Pi Imager.
3. These instructions are written for **Raspberry Pi Imager v2.0.3** and **Debian Trixie**
3. **Select your Raspberry Pi device** – here an RPi4



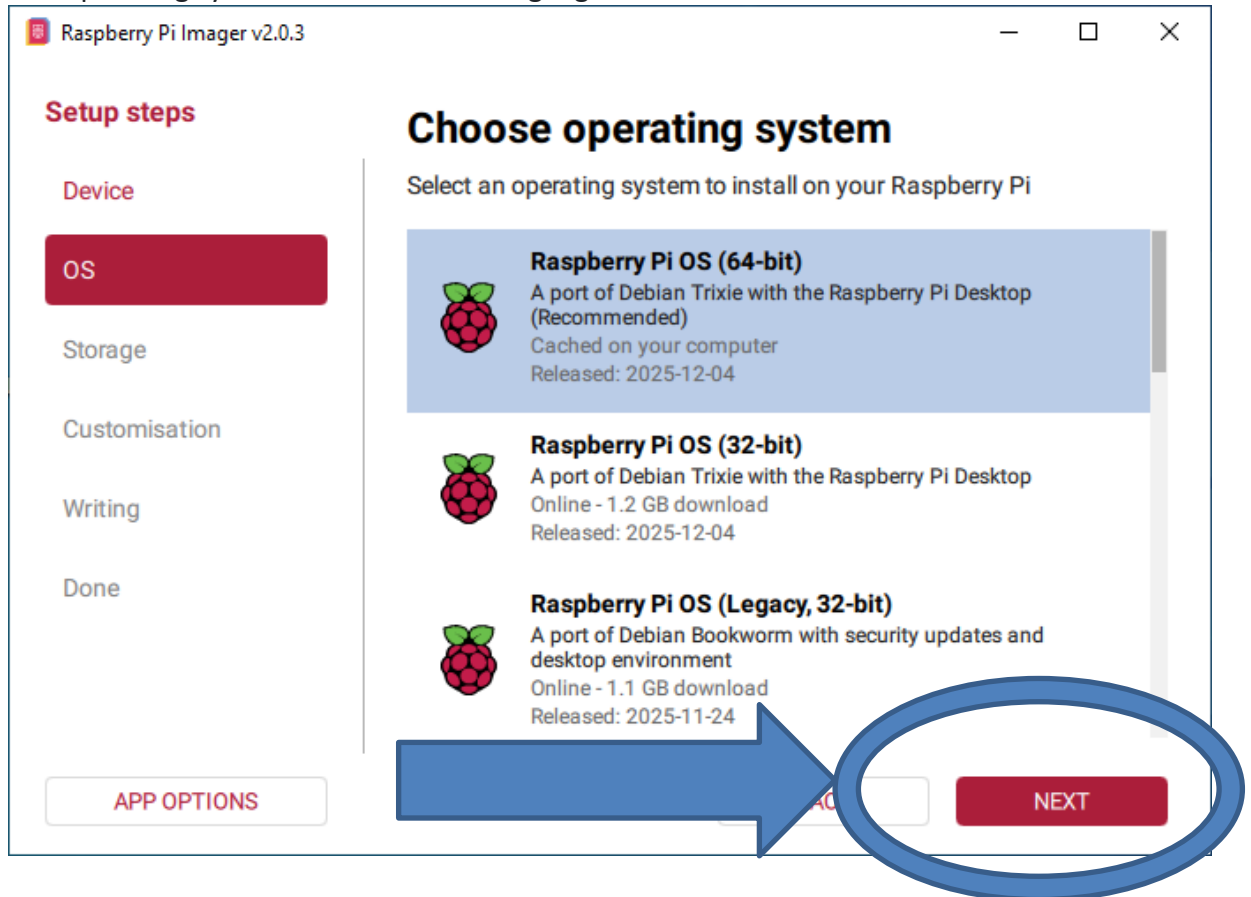
4. The device selected will be highlighter. Then select **NEXT**.



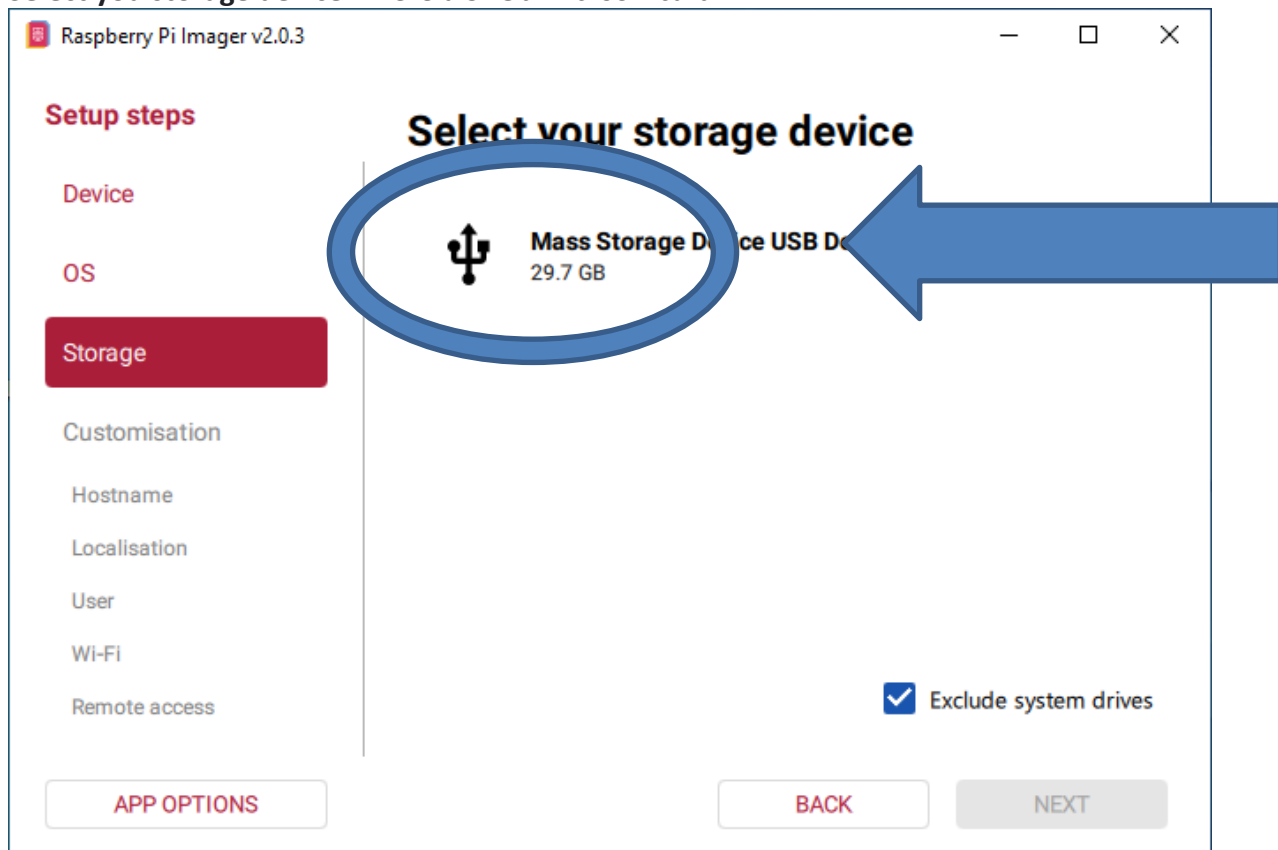
5. **Select the operating system** – here Raspberry Pi OS (64-bit).



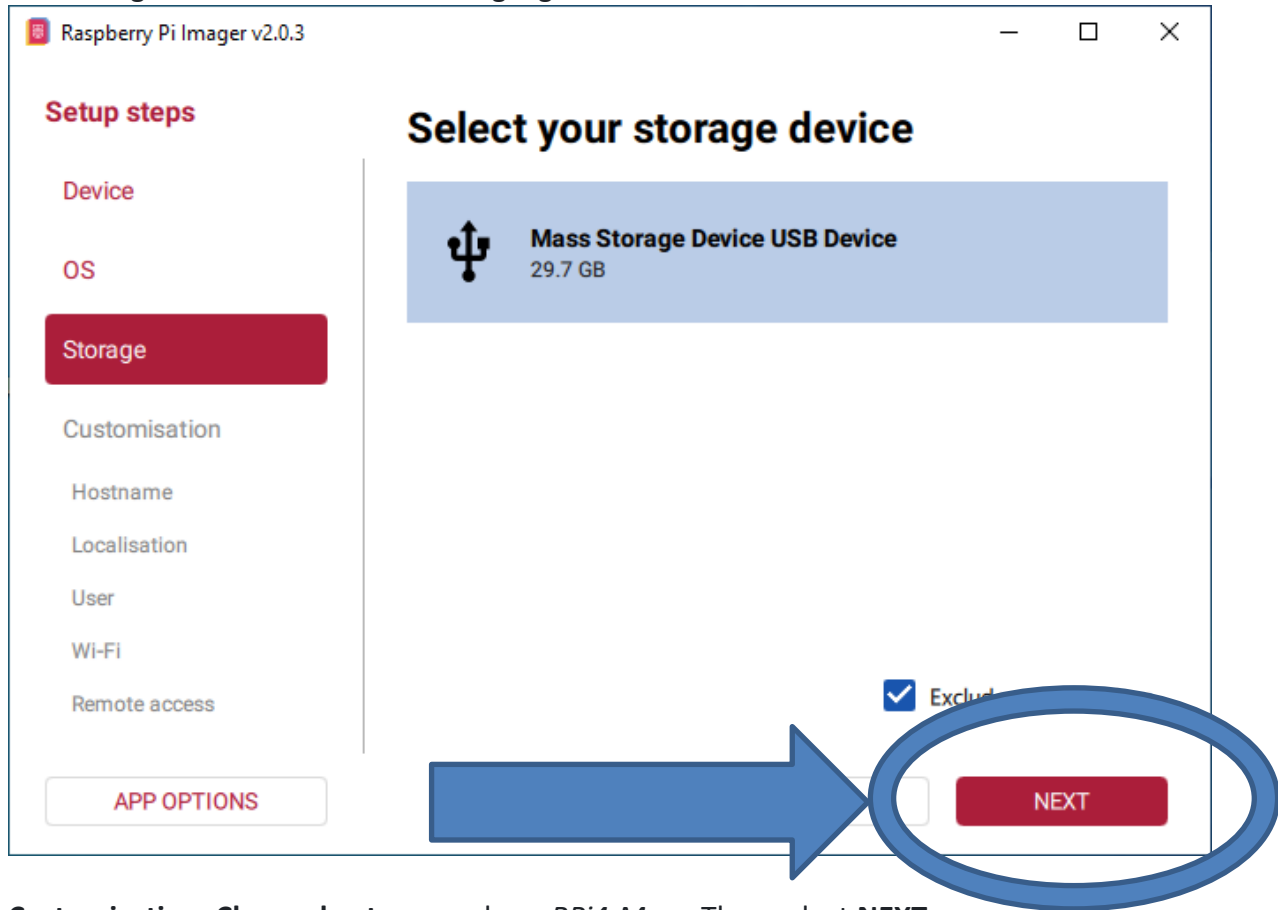
6. The operating system selected will be highlighted – then select **NEXT**.



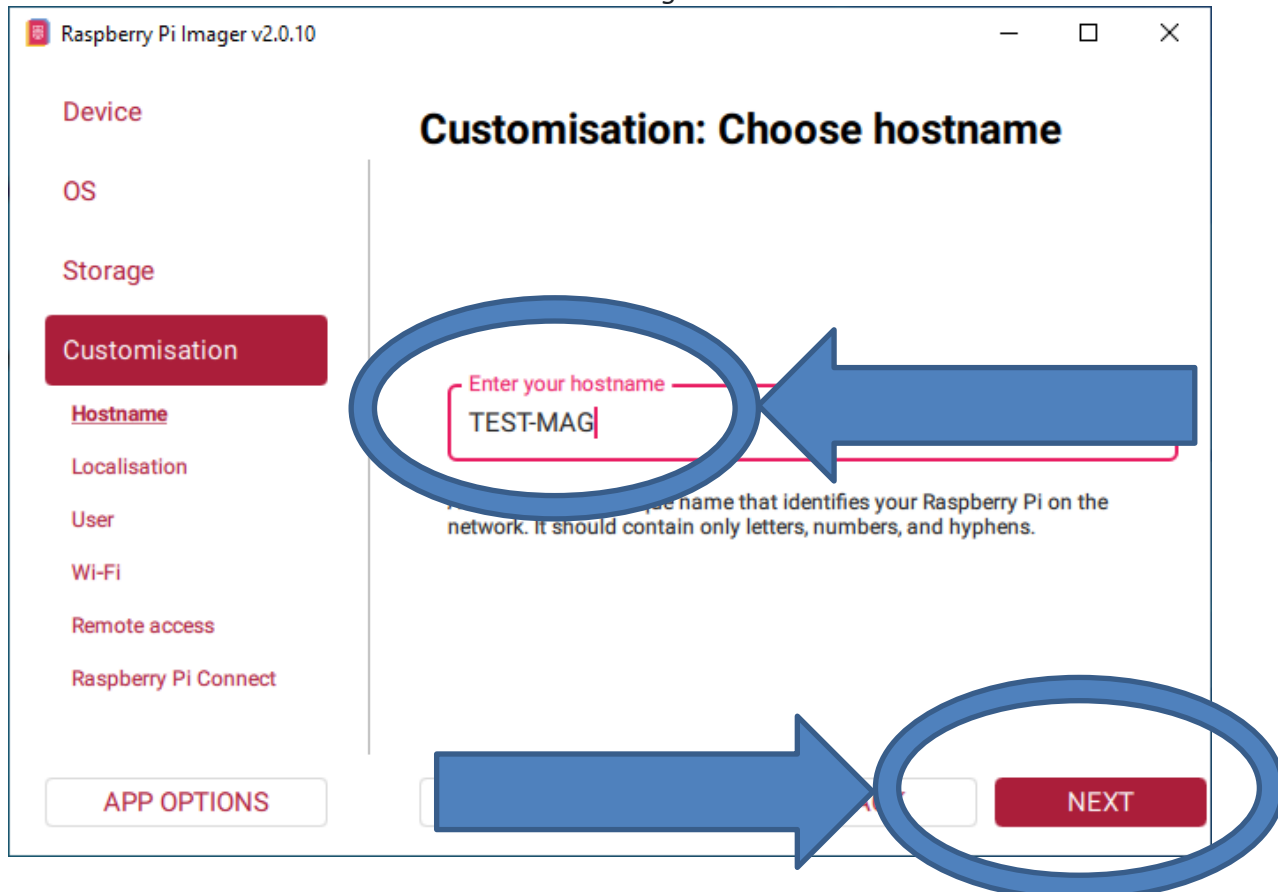
7. Select your storage device – here a 32Gb microSD card.



8. The storage device selected will be highlighted – then select **NEXT**.



9. **Customisation: Choose hostname** – here *RPi4-Mag*. Then select **NEXT**



10. **Customisation: Localisation** – change for your specific location. Then select **NEXT**

Raspberry Pi Imager v2.0.3

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

APP OPTIONS

Capital city: London (United Kingdom)

Time zone: Europe/London

Keyboard layout: gb

NEXT

11. **Customisation: Choose username** – leave as **pi**, create password. Then select **NEXT**

Raspberry Pi Imager v2.0.3

Customisation: Choose username

Create a user account for your Raspberry Pi

Storage

Customisation

Hostname

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Enter your username

pi

Password: Saved (hidden) — leave blank to keep

Confirm password: Re-enter to change password

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

NEXT

12. **Customisation: Choose Wi-Fi** – enter your Wi-Fi details. Then select **NEXT**.

Raspberry Pi Imager v2.0.3

Customisation: Choose Wi-Fi

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

SECURE NETWORK OPEN NETWORK

SSID: Your_wireless_name

Password: Your_wireless_password

Confirm password:

☐ Hidden SSID

NEXT

13. **Customisation: SSH authentication** – enable SSH. Then select **NEXT**.

Raspberry Pi Imager v2.0.3

Customisation: SSH authentication

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Configure SSH access

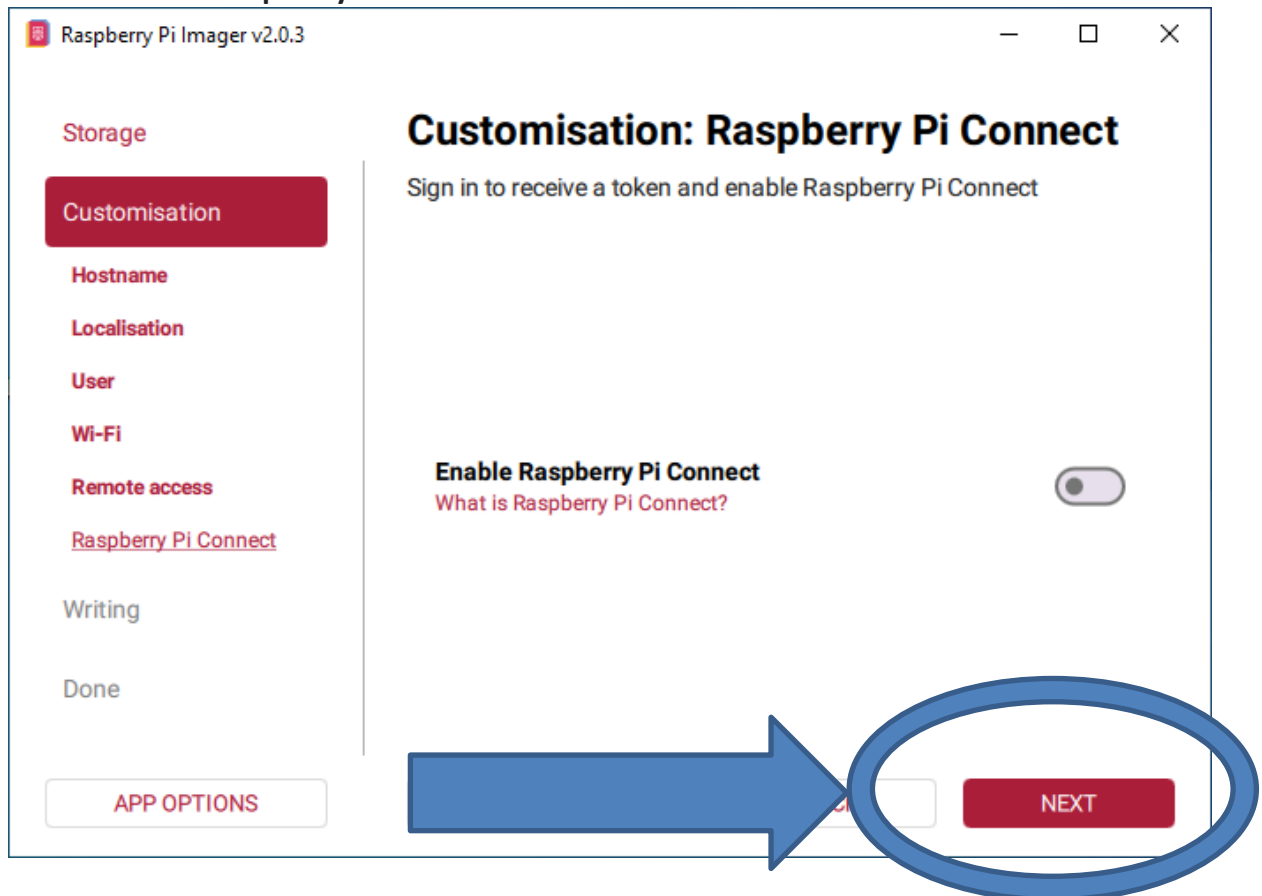
Authentication mechanism:

☒ Use password

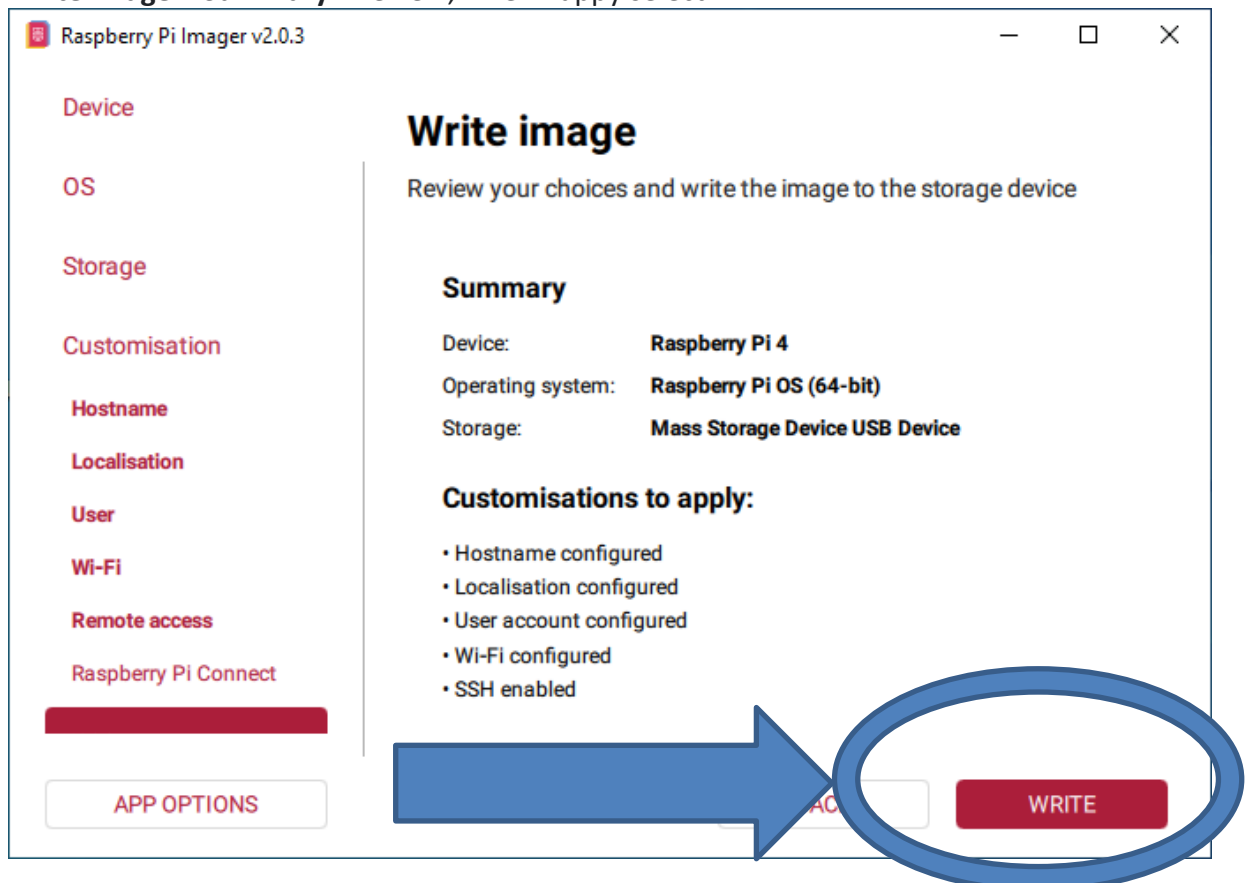
☐ Use public key authentication

NEXT

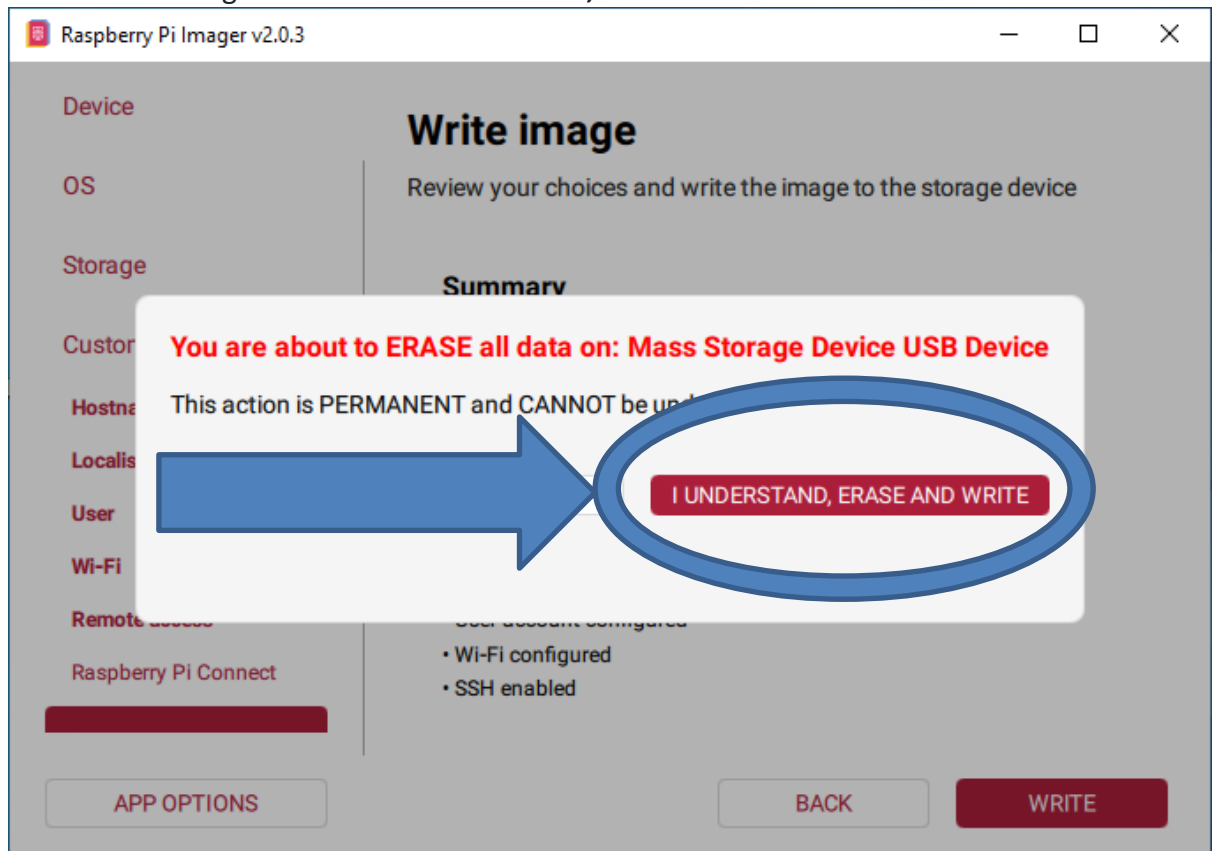
14. **Customisation: Raspberry Pi Connect** – enable if wanted. Then select **NEXT**.



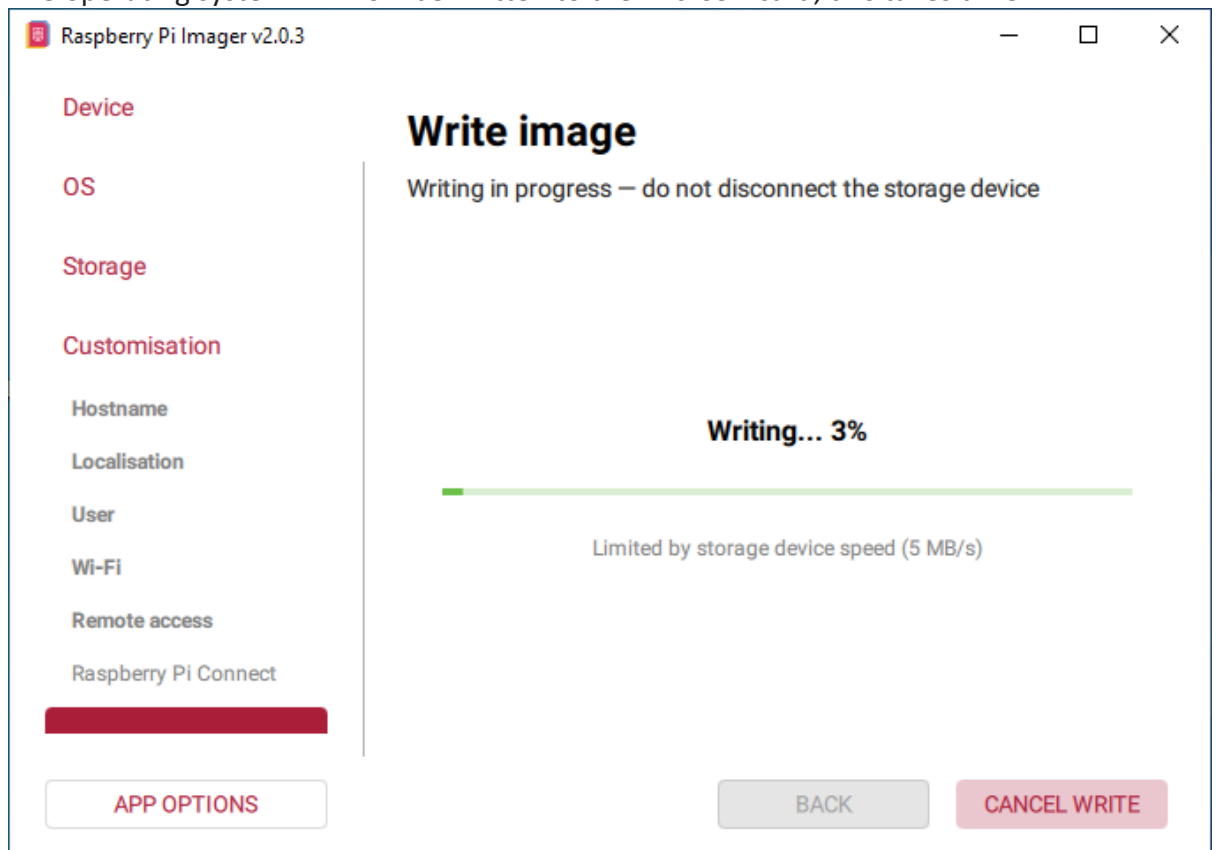
15. **Write image – Summary** – review, when happy select **WRITE**.



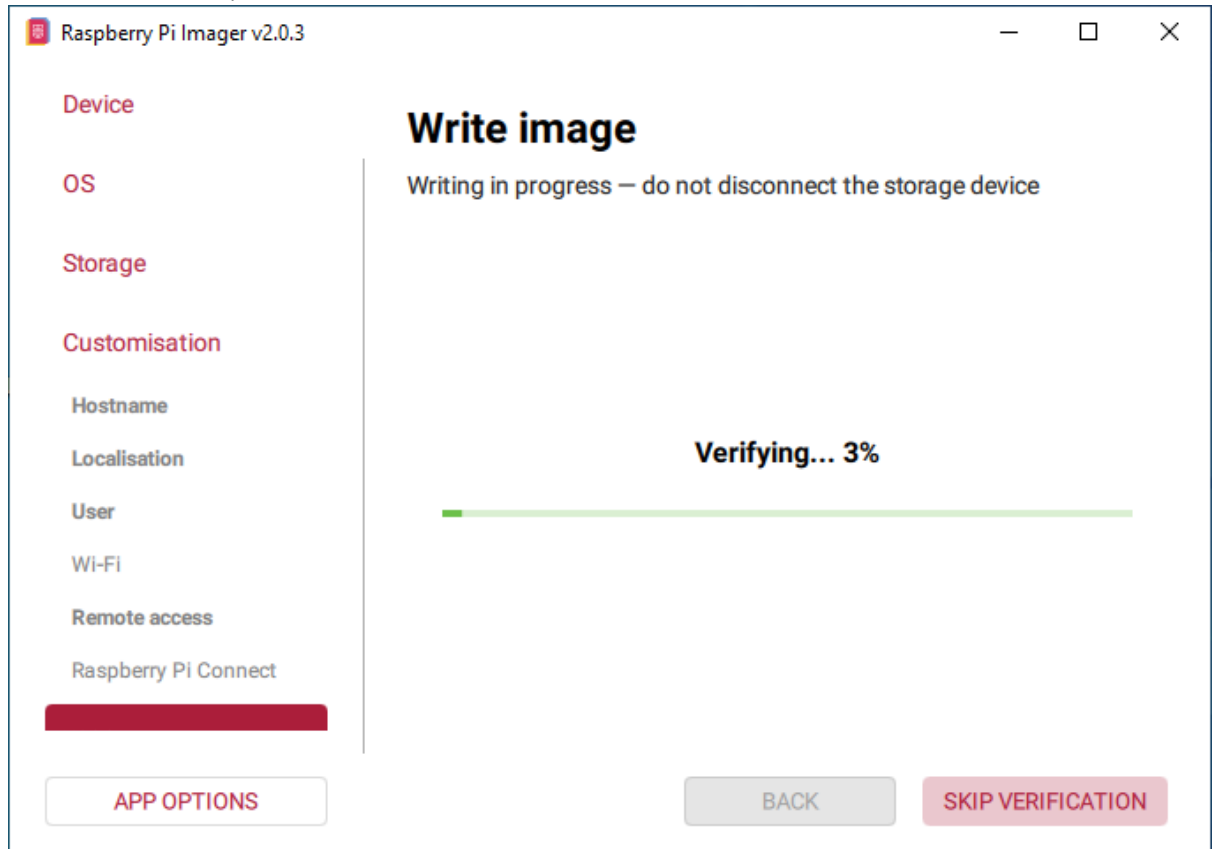
16. Read the warning and select **I UNDERSTAND, ERASE AND WRITE**.



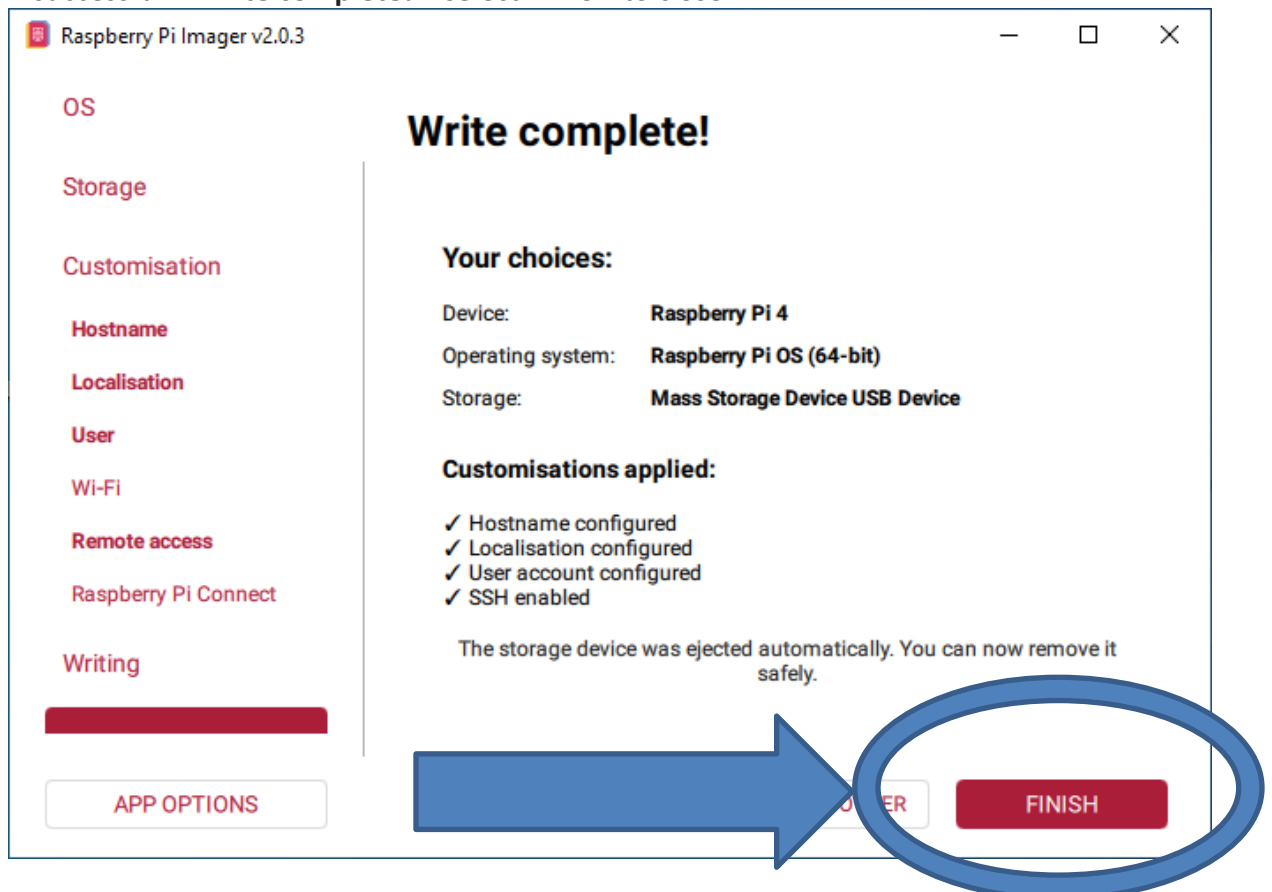
17. The operating system will now be written to the microSD card, this takes time...



18. and then verified, this take a bit more time...



19. If successful – **Write complete!** – select **FINISH** to close.

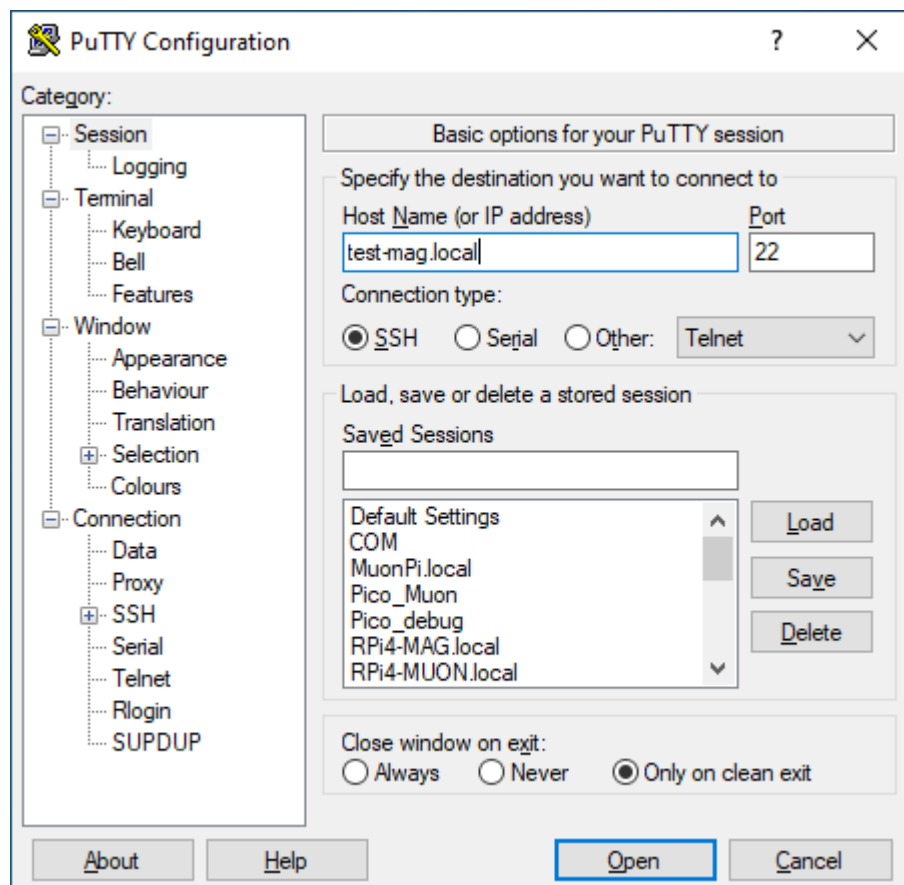


RPi4 first boot and initial configuration

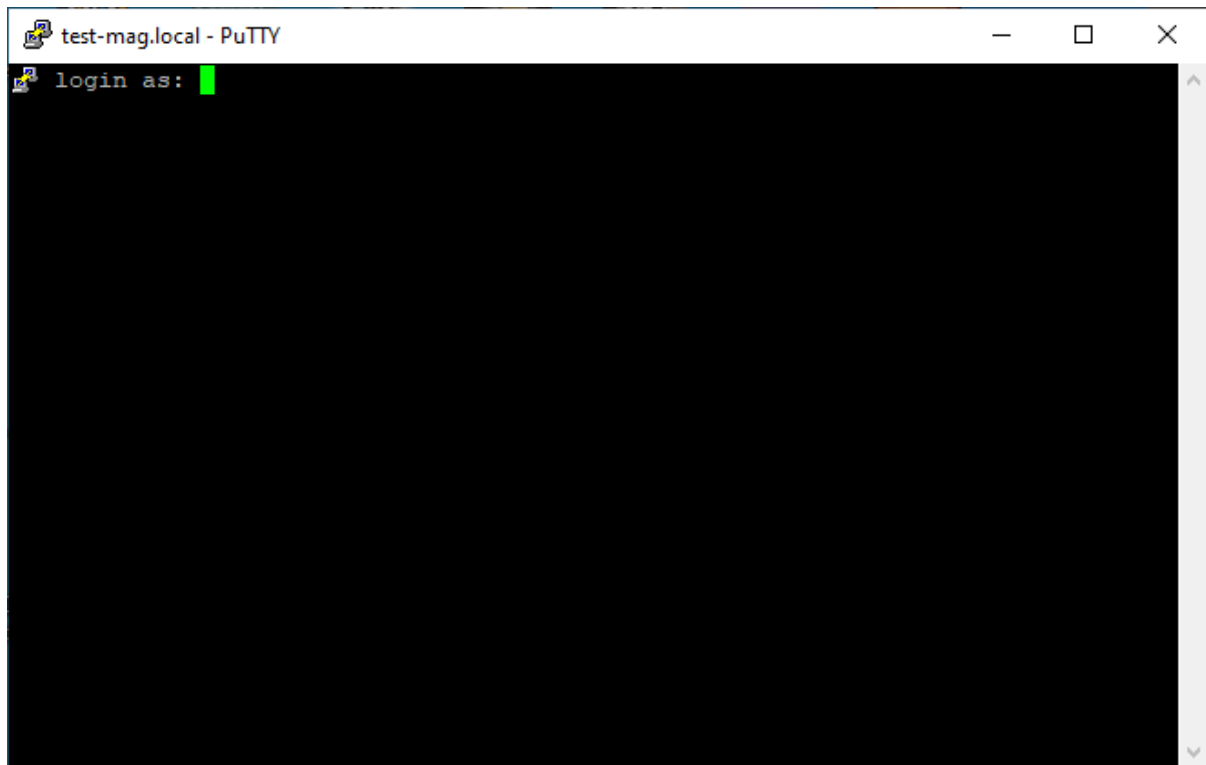
Put the microSD card you have just created into your RPi4, connect the RPi to your LAN if necessary, insert the power cable and turn on.

Using PuTTY (<https://www.putty.org/>)

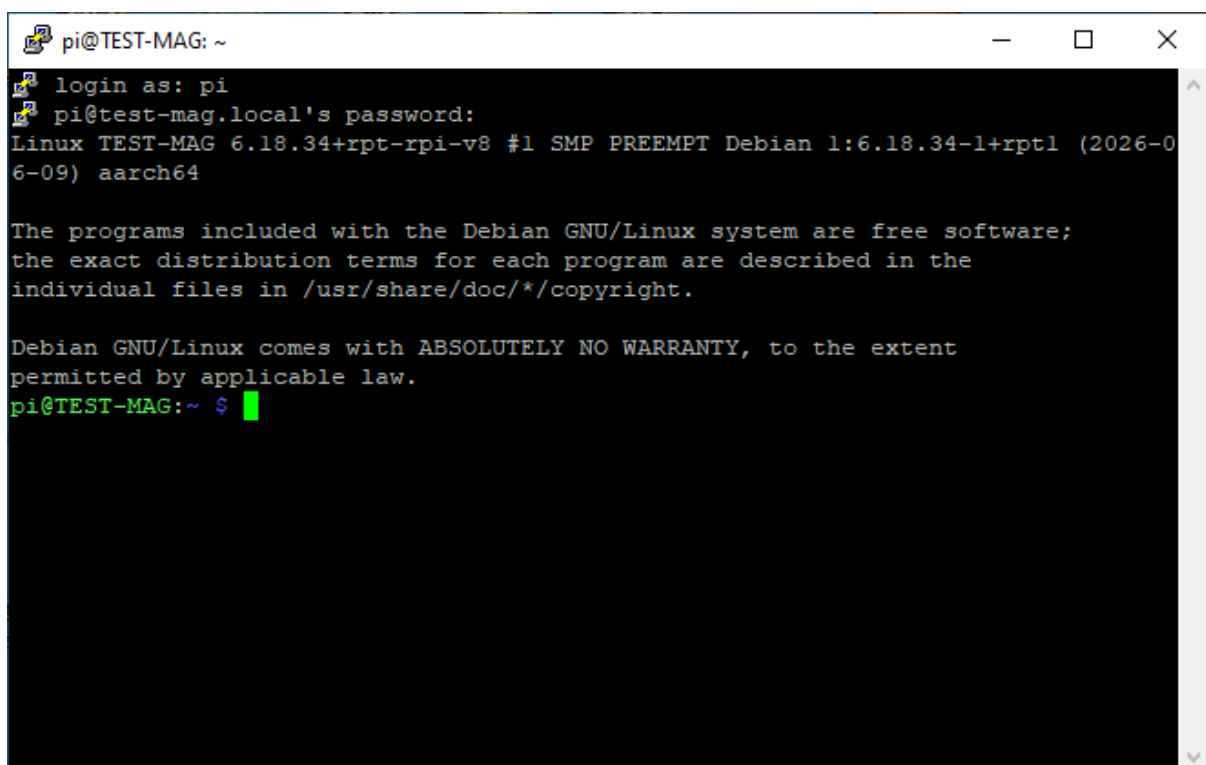
1. In **Host Name (or IP address)**, type the host name of your RPi that you wrote down earlier... and select **Open**



2. You will be presented with a PuTTY Security Alert - select **Accept**. You will now have a PuTTY terminal window.

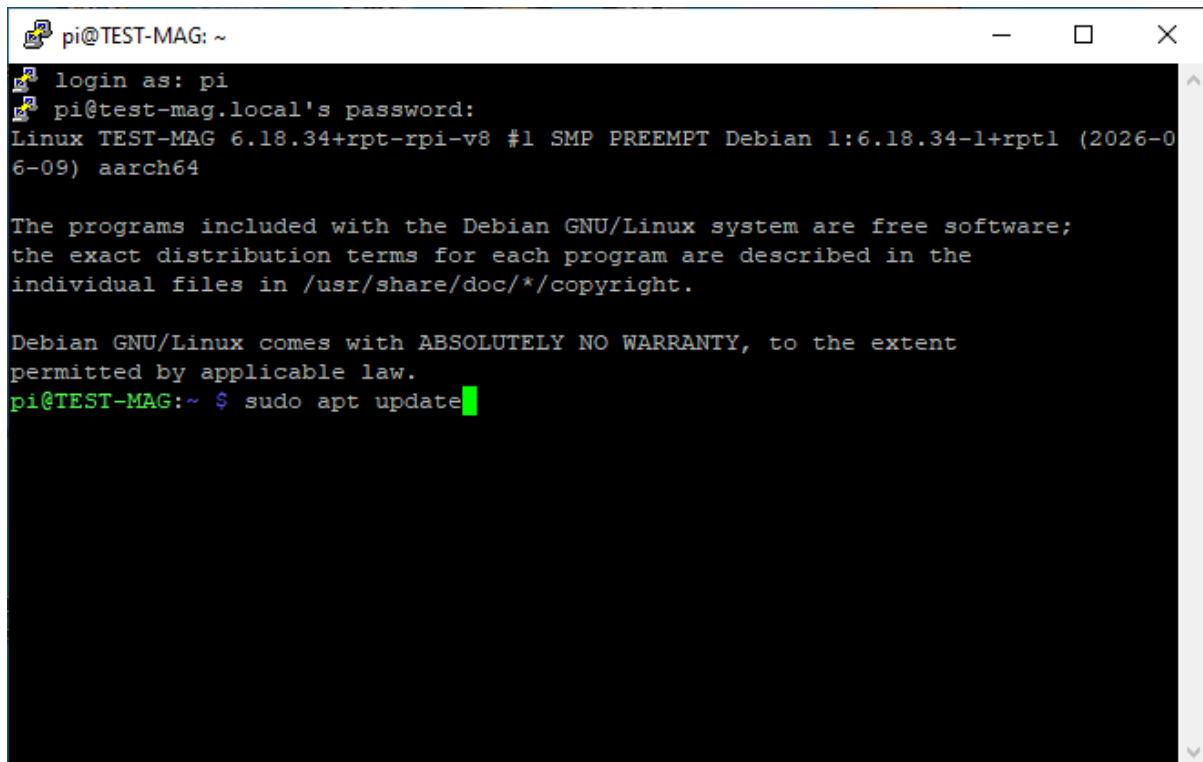


3. Login as **pi** and enter your **password**



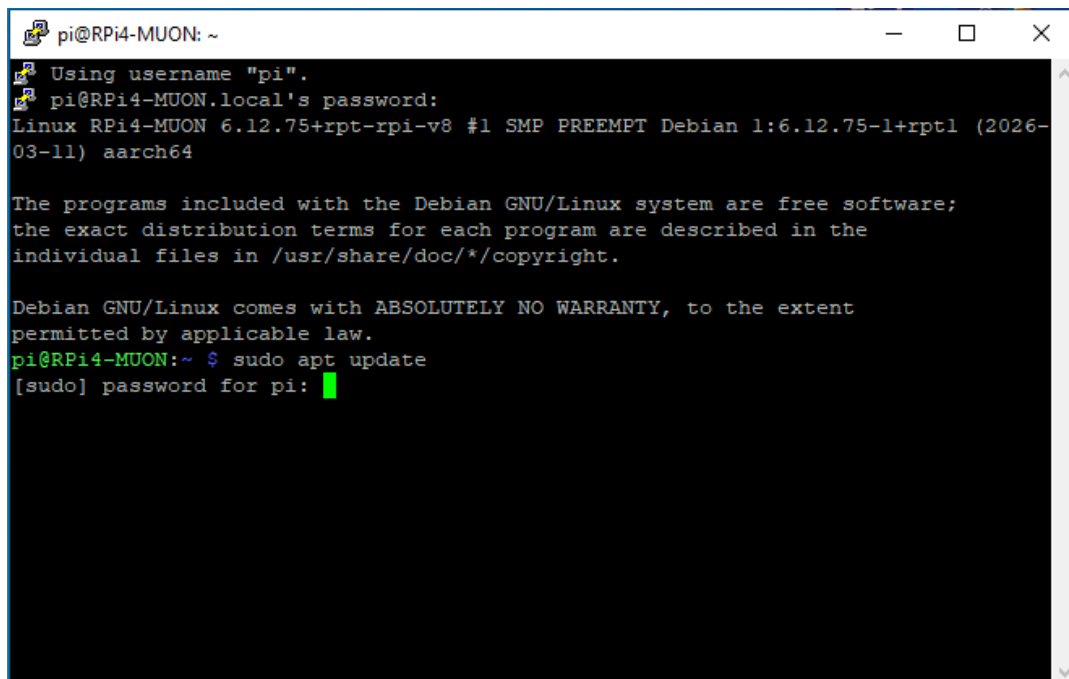
4. Now type the command below and press **enter**. This will update the RPi OS

sudo apt update



```
pi@TEST-MAG: ~  
login as: pi  
pi@test-mag.local's password:  
Linux TEST-MAG 6.18.34+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.18.34-1+rpt1 (2026-06-09) aarch64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
pi@TEST-MAG:~ $ sudo apt update
```

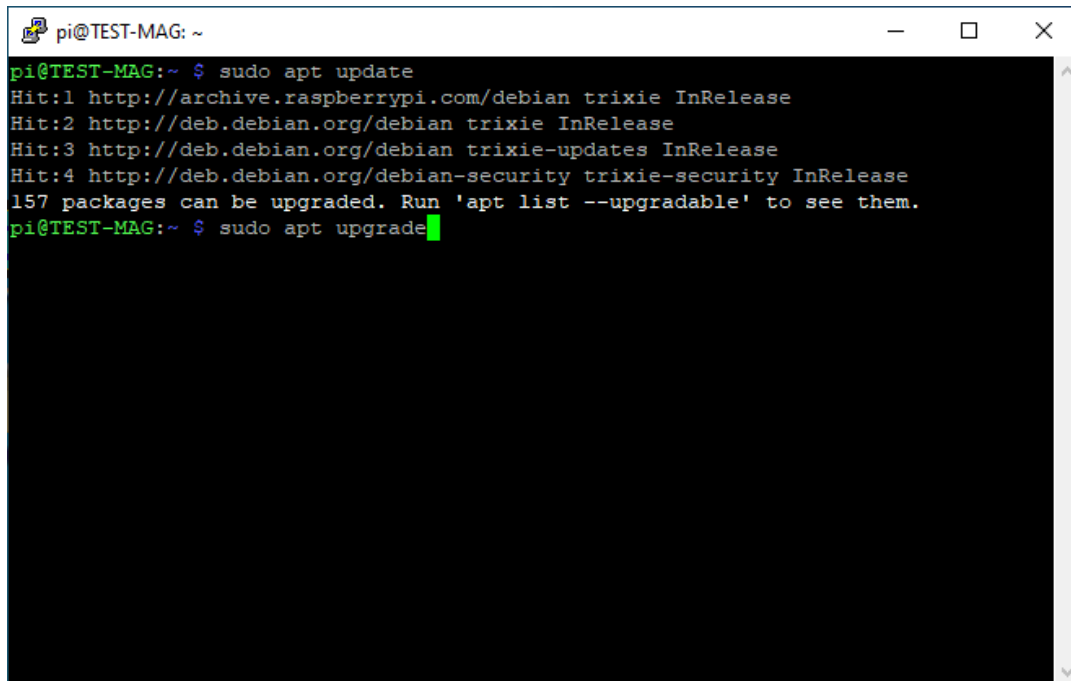
5. If you are asked for your password, enter your **password** and press enter.



```
pi@RPi4-MUON: ~  
Using username "pi".  
pi@RPi4-MUON.local's password:  
Linux RPi4-MUON 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
pi@RPi4-MUON:~ $ sudo apt update  
[sudo] password for pi:
```


- Now type the command below and press **enter**. This will update any preinstalled software packages. You will be asked to type **y** and press **enter** to proceed.

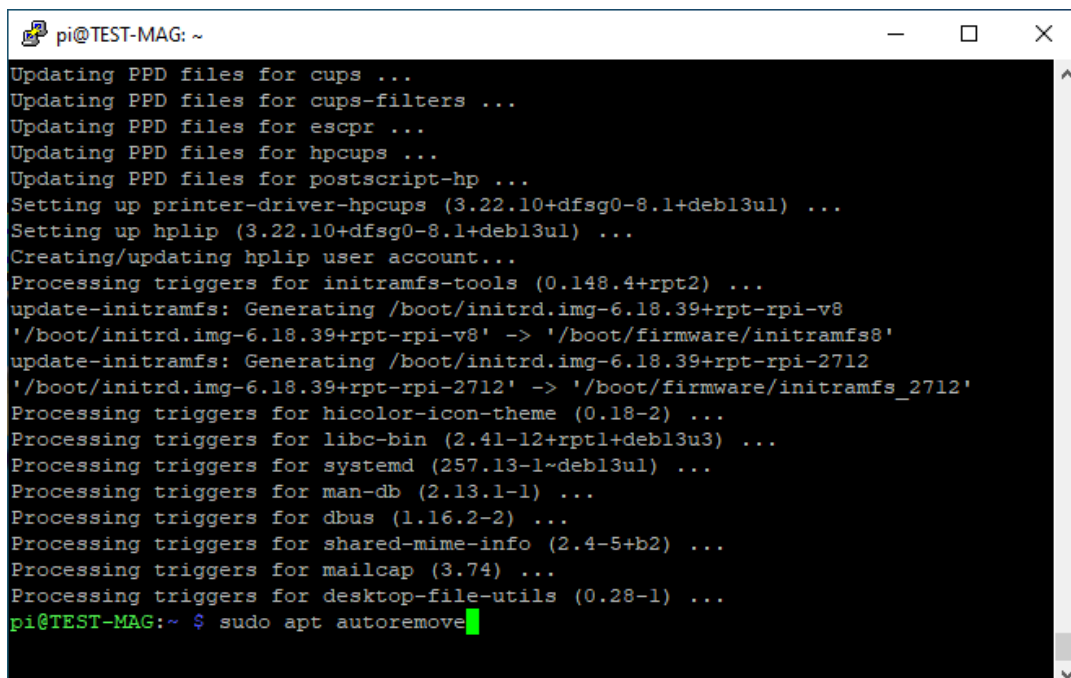
sudo apt upgrade



```
pi@TEST-MAG: ~  
pi@TEST-MAG:~ $ sudo apt update  
Hit:1 http://archive.raspberrypi.com/debian trixie InRelease  
Hit:2 http://deb.debian.org/debian trixie InRelease  
Hit:3 http://deb.debian.org/debian trixie-updates InRelease  
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease  
157 packages can be upgraded. Run 'apt list --upgradable' to see them.  
pi@TEST-MAG:~ $ sudo apt upgrade
```

- You may be advised that you can remove **no longer required** packages. You can type the command below and press **enter**. Again, you will be asked to type **y** and press **enter** to proceed.

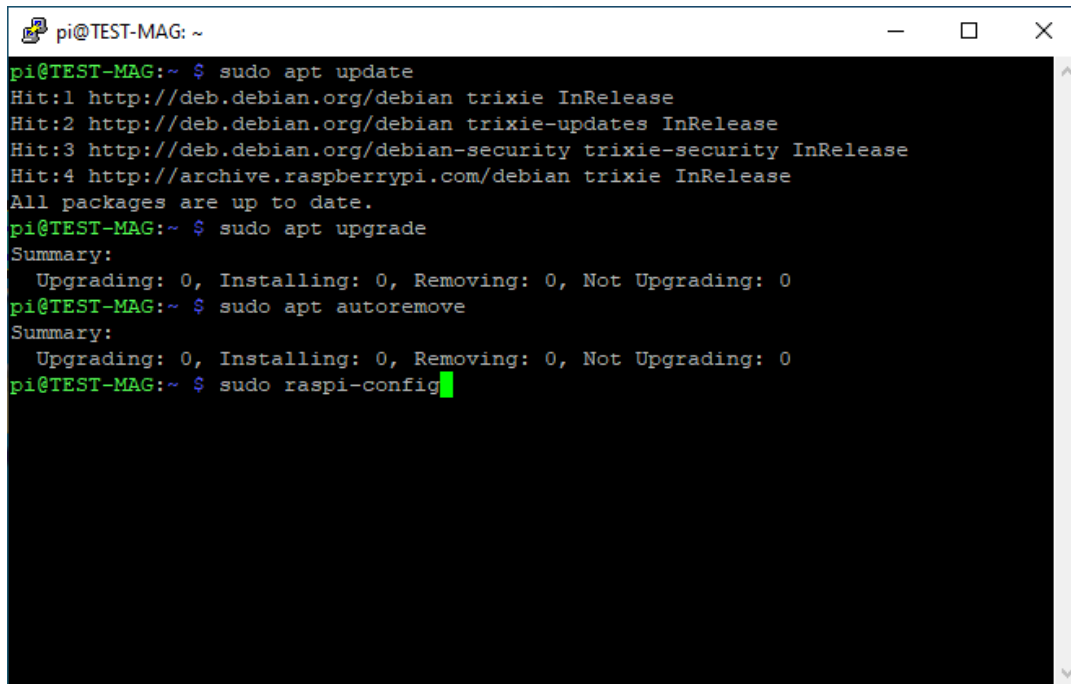
sudo apt autoremove



```
pi@TEST-MAG: ~  
Updating PPD files for cups ...  
Updating PPD files for cups-filters ...  
Updating PPD files for escpr ...  
Updating PPD files for hpcups ...  
Updating PPD files for postscript-hp ...  
Setting up printer-driver-hpcups (3.22.10+dfsg0-8.1+deb13ul) ...  
Setting up hplip (3.22.10+dfsg0-8.1+deb13ul) ...  
Creating/updating hplip user account...  
Processing triggers for initramfs-tools (0.148.4+rpt2) ...  
update-initramfs: Generating /boot/initrd.img-6.18.39+rpt-rpi-v8  
'/boot/initrd.img-6.18.39+rpt-rpi-v8' -> '/boot/firmware/initramfs8'  
update-initramfs: Generating /boot/initrd.img-6.18.39+rpt-rpi-2712  
'/boot/initrd.img-6.18.39+rpt-rpi-2712' -> '/boot/firmware/initramfs_2712'  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for libc-bin (2.41-12+rpt1+deb13u3) ...  
Processing triggers for systemd (257.13-1~deb13ul) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for dbus (1.16.2-2) ...  
Processing triggers for shared-mime-info (2.4-5+b2) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
pi@TEST-MAG:~ $ sudo apt autoremove
```

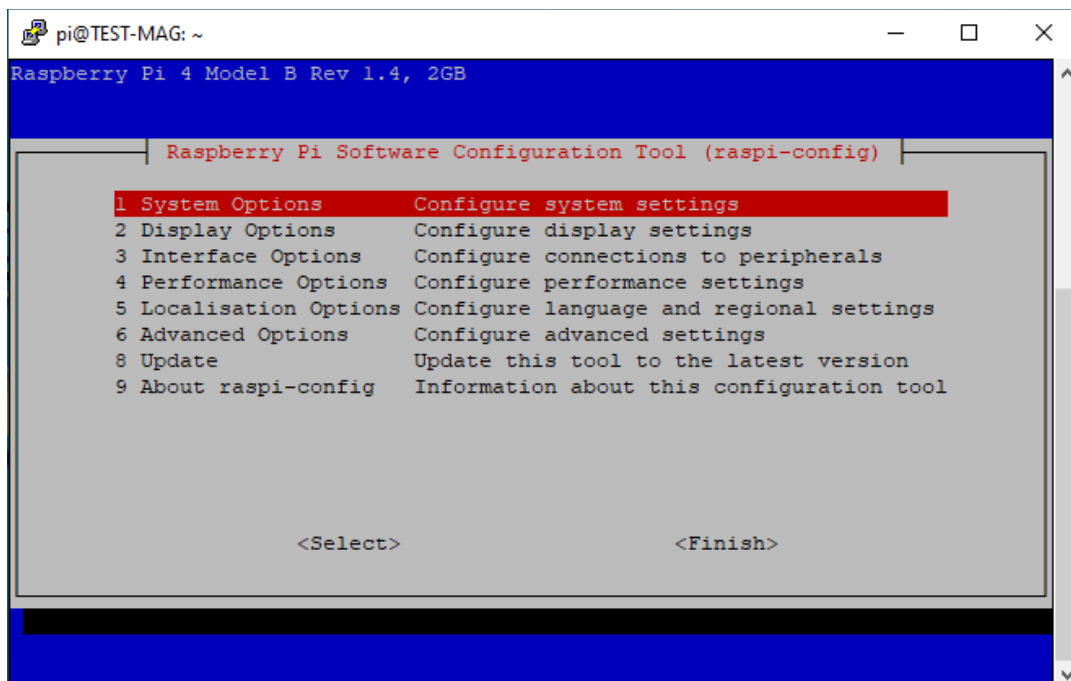
8. We can now set up VNC to access the RPi from our desktop PC. Type the command below and press **enter**. We will be presented with the RPi configuration tool.

sudo raspi-config



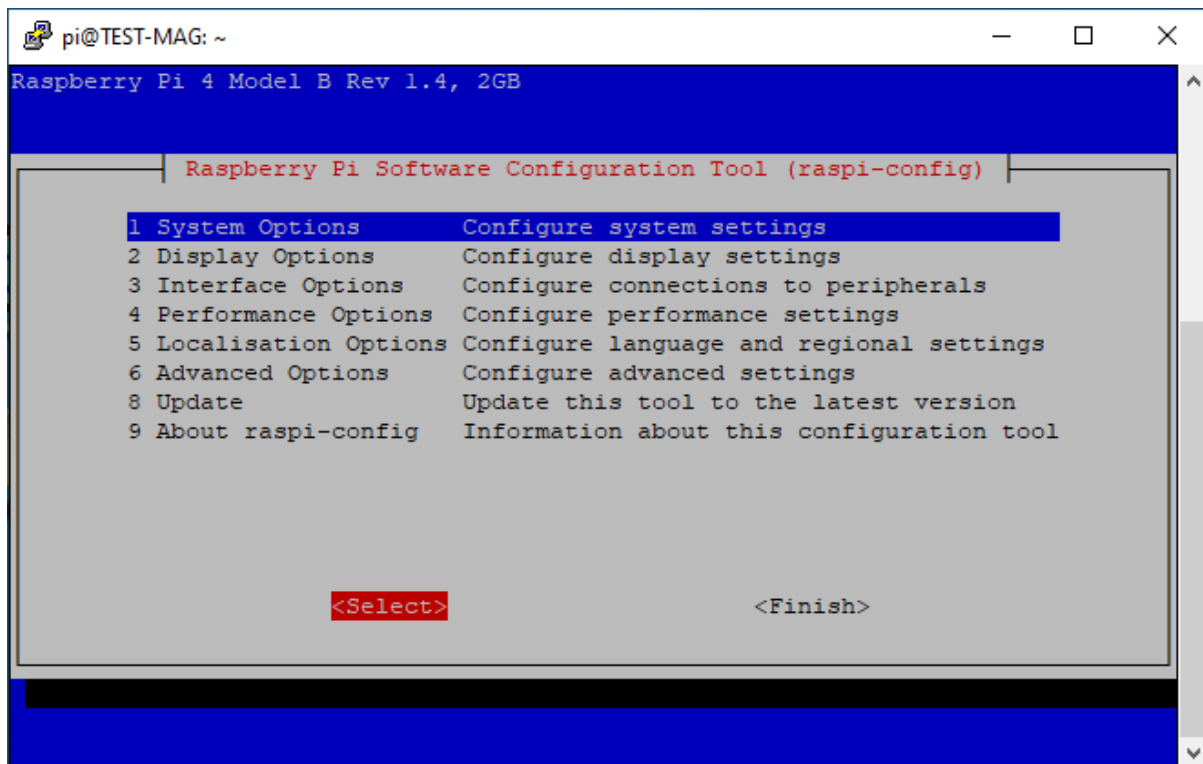
```
pi@TEST-MAG: ~  
pi@TEST-MAG:~ $ sudo apt update  
Hit:1 http://deb.debian.org/debian trixie InRelease  
Hit:2 http://deb.debian.org/debian trixie-updates InRelease  
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease  
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease  
All packages are up to date.  
pi@TEST-MAG:~ $ sudo apt upgrade  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
pi@TEST-MAG:~ $ sudo apt autoremove  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
pi@TEST-MAG:~ $ sudo raspi-config
```

9. If you had to enter a password in step 4/5 then Select **1 System Options**, else move to step 15.

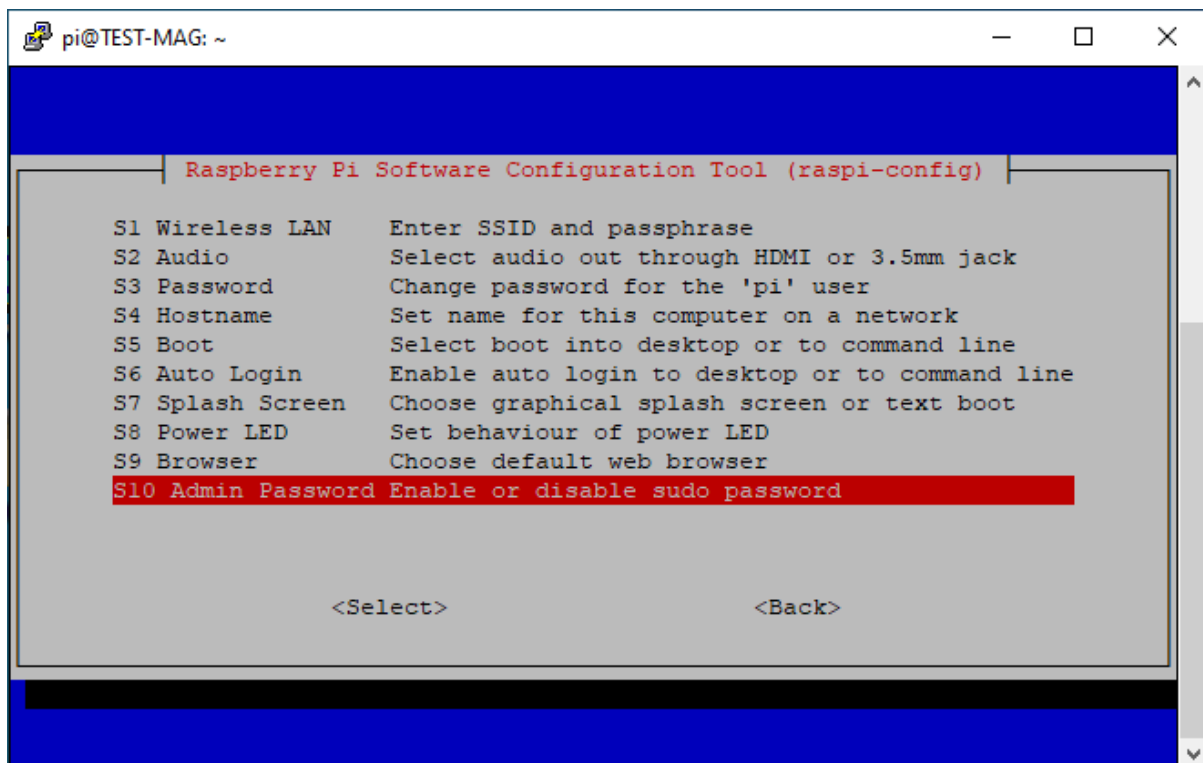


```
pi@TEST-MAG: ~  
Raspberry Pi 4 Model B Rev 1.4, 2GB  
Raspberry Pi Software Configuration Tool (raspi-config)  
1 System Options      Configure system settings  
2 Display Options     Configure display settings  
3 Interface Options   Configure connections to peripherals  
4 Performance Options Configure performance settings  
5 Localisation Options Configure language and regional settings  
6 Advanced Options    Configure advanced settings  
8 Update              Update this tool to the latest version  
9 About raspi-config  Information about this configuration tool  
  
<Select>              <Finish>
```

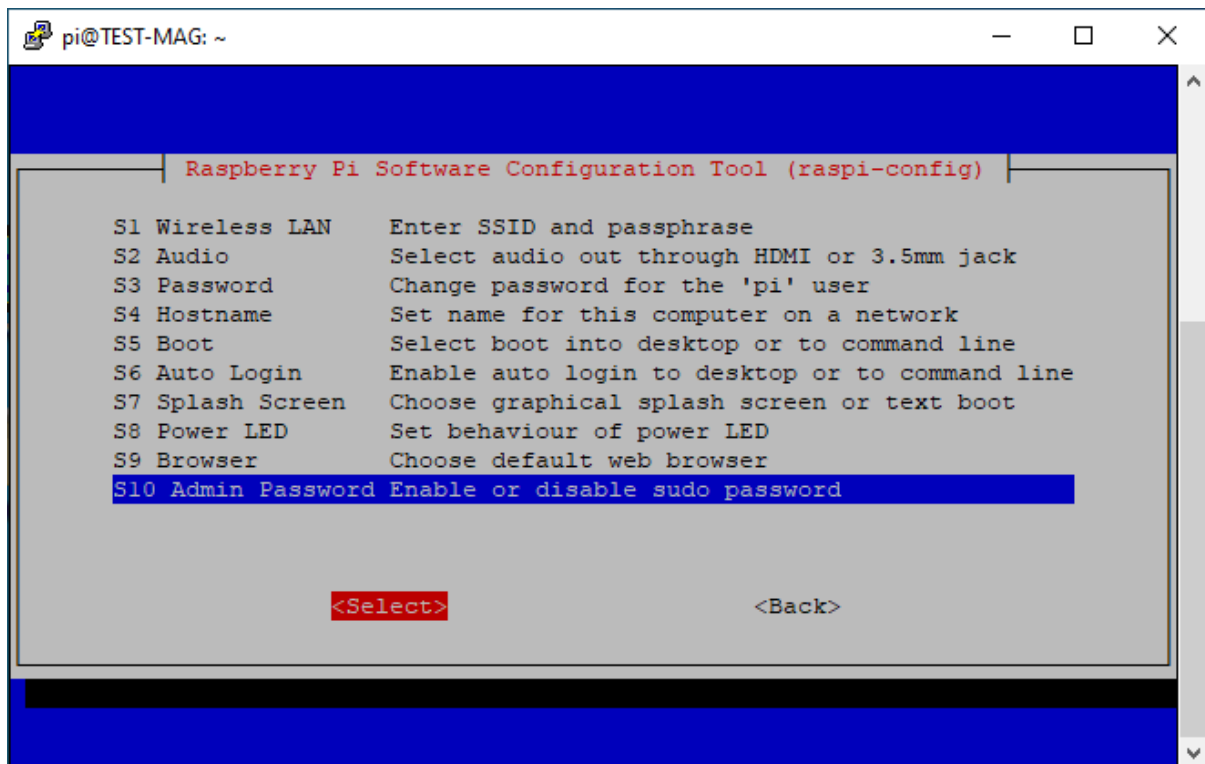
10. Navigate to **Select** and press **enter**.



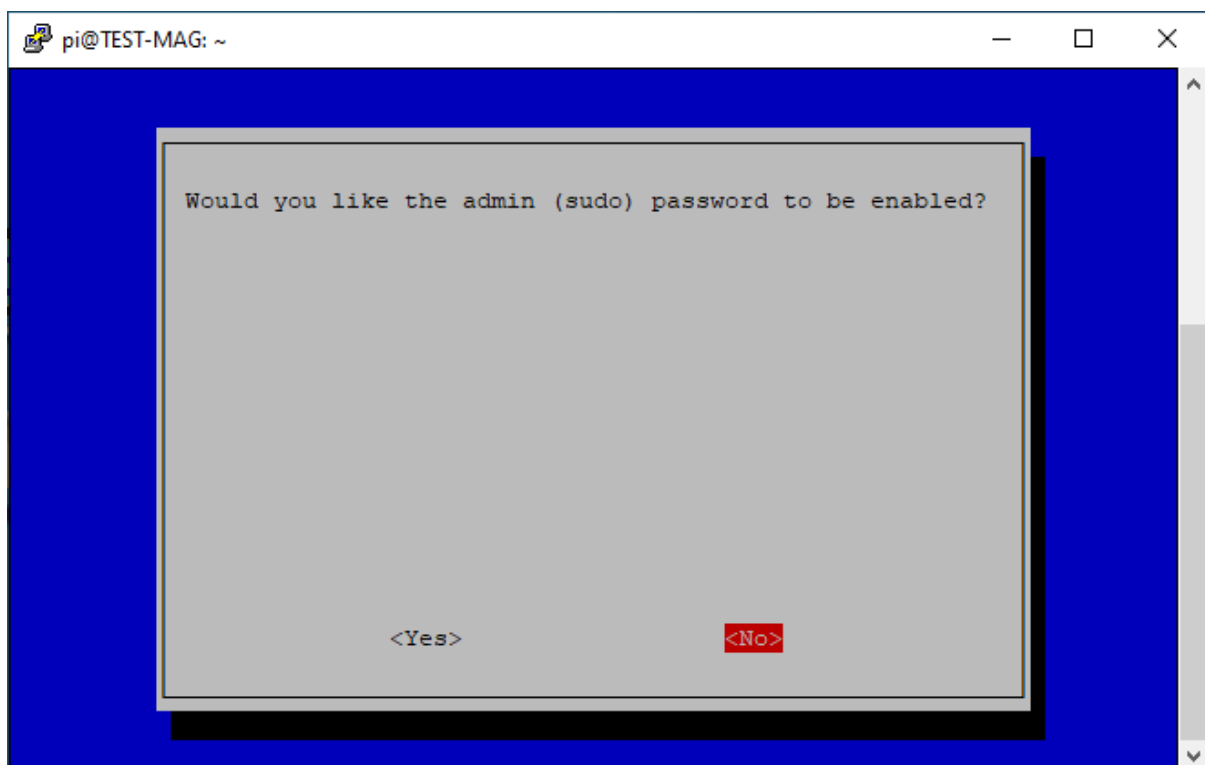
11. Select **S10 Admin Password**



12. Navigate to **Select** and press **enter**.



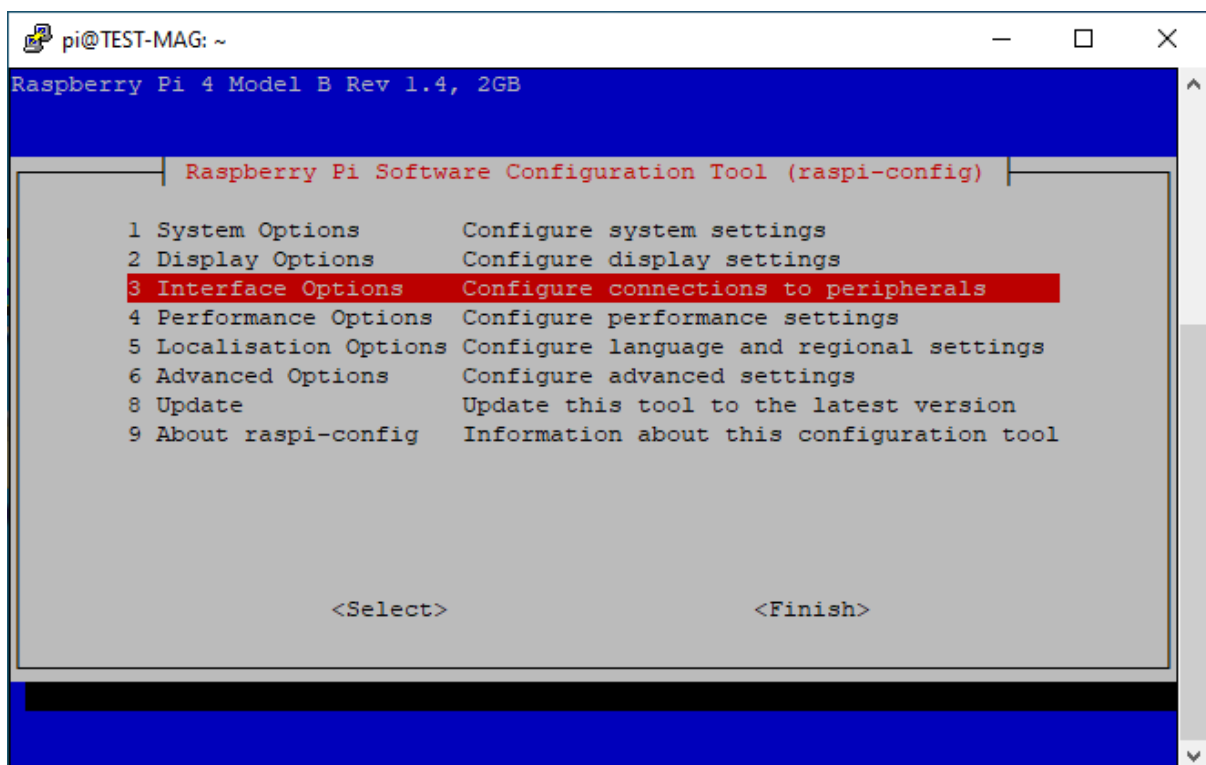
13. Select **No** and press **enter** to disable admin (sudo) password.



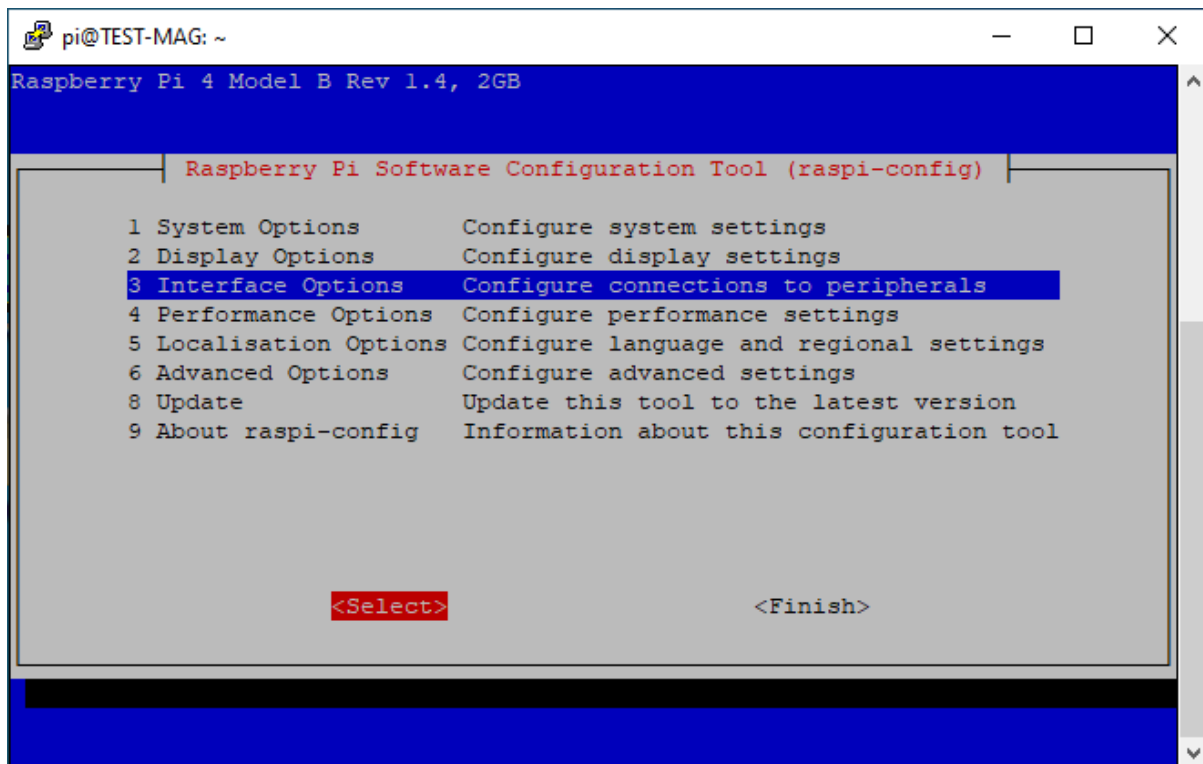
14. You will be informed that the admin password is disabled, press **enter**.



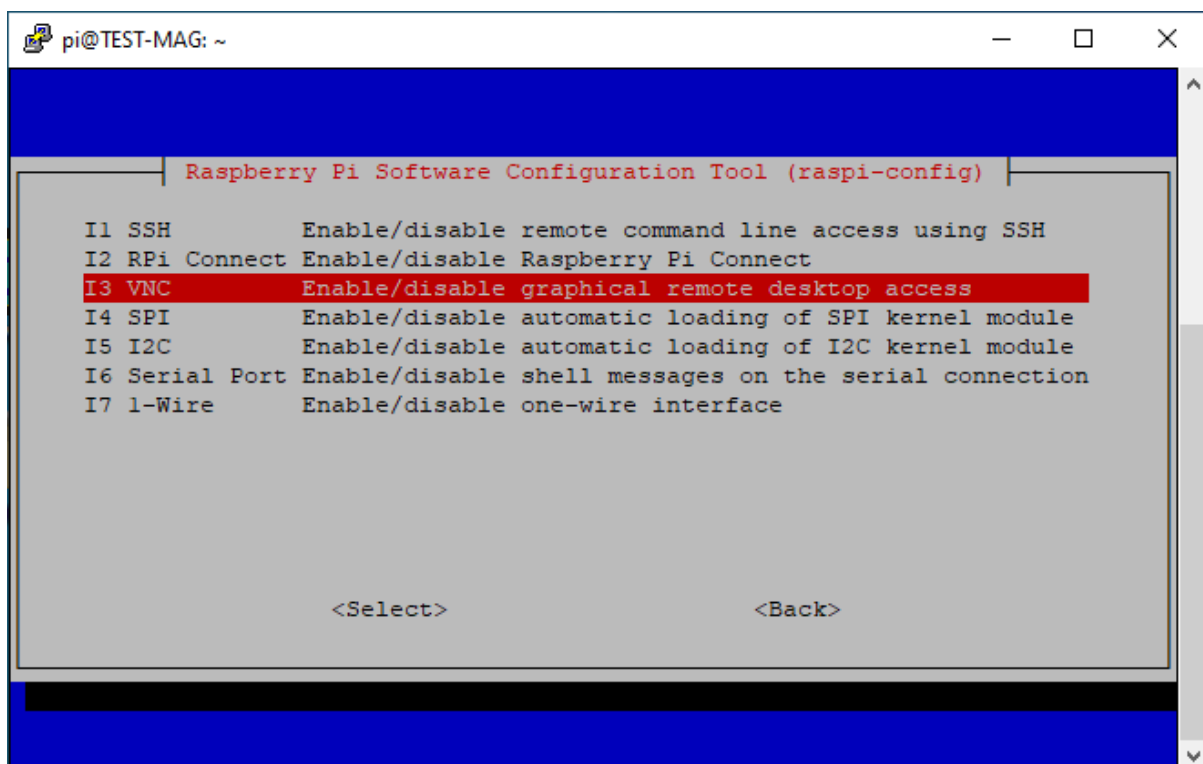
15. Select **I3 Interface Options**



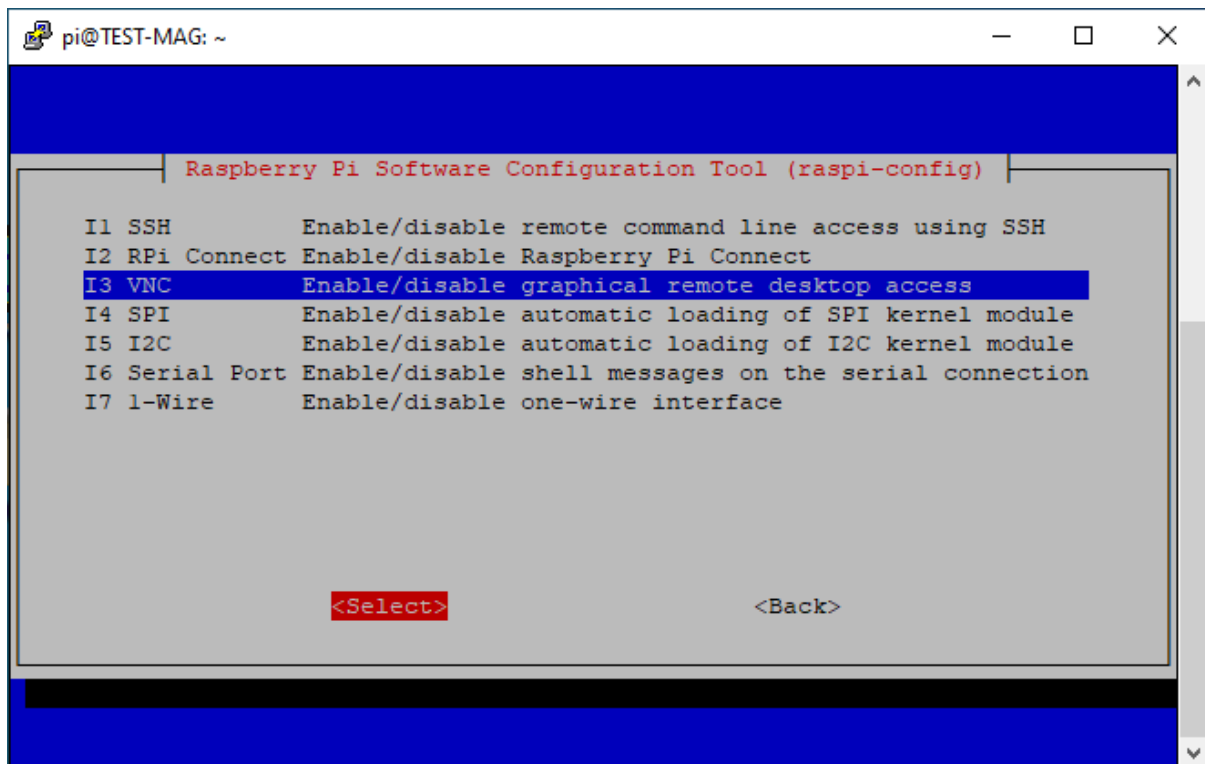
16. Navigate to **Select** and press **enter**.



17. Now navigate down to **I3 VNC**.



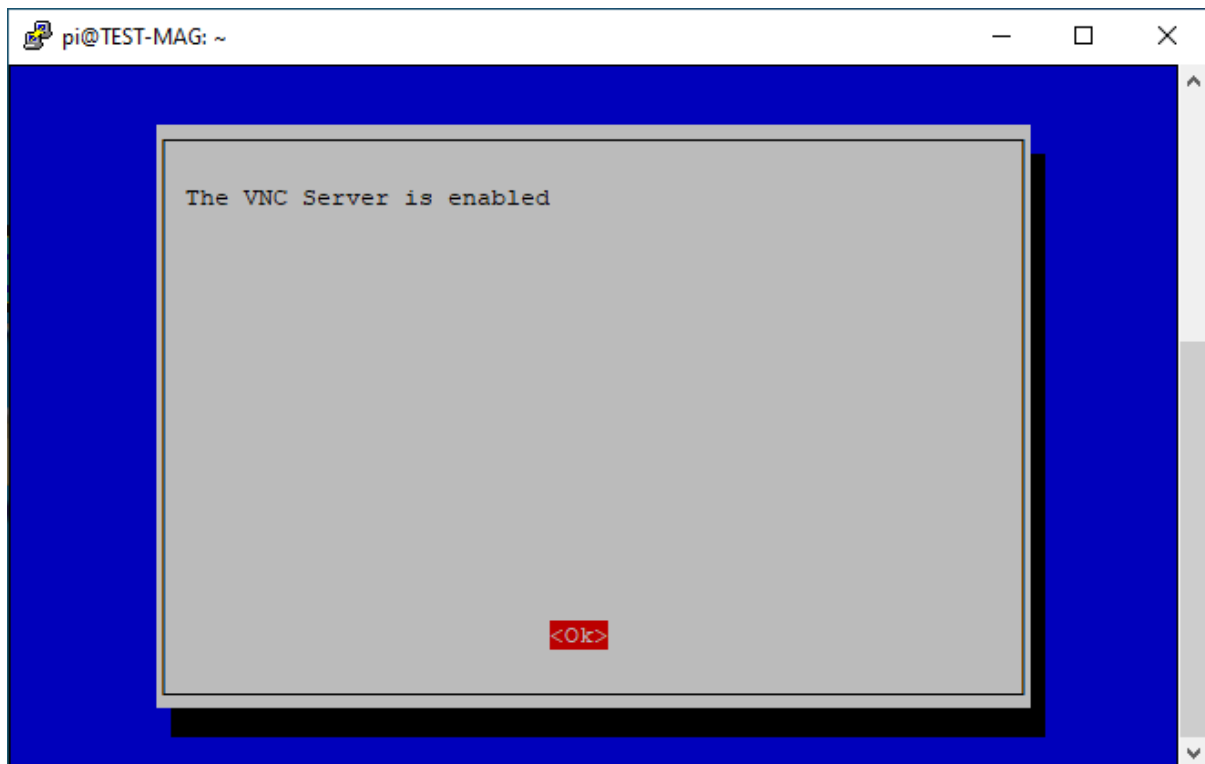
18. Navigate to **Select** and press **enter**.



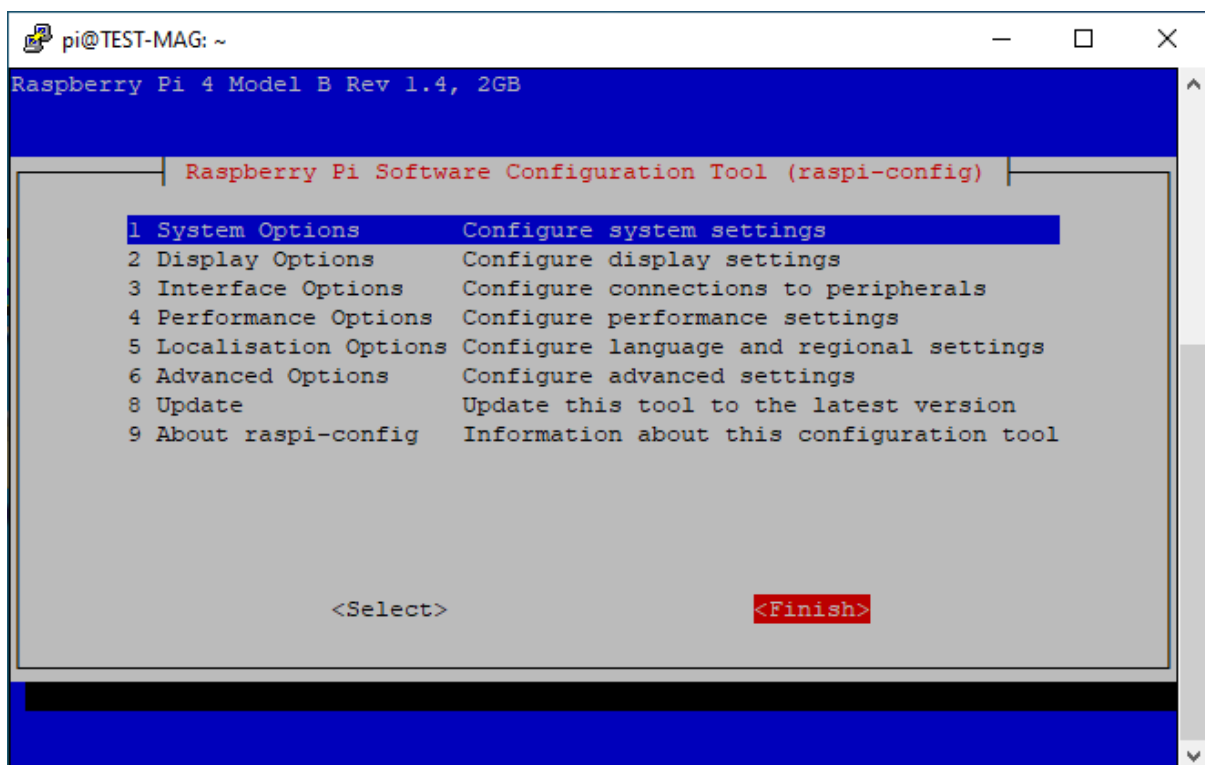
19. Select **Yes** and press **enter** to enable the VNC Server.



20. VNC server will now be enabled, press **enter**.

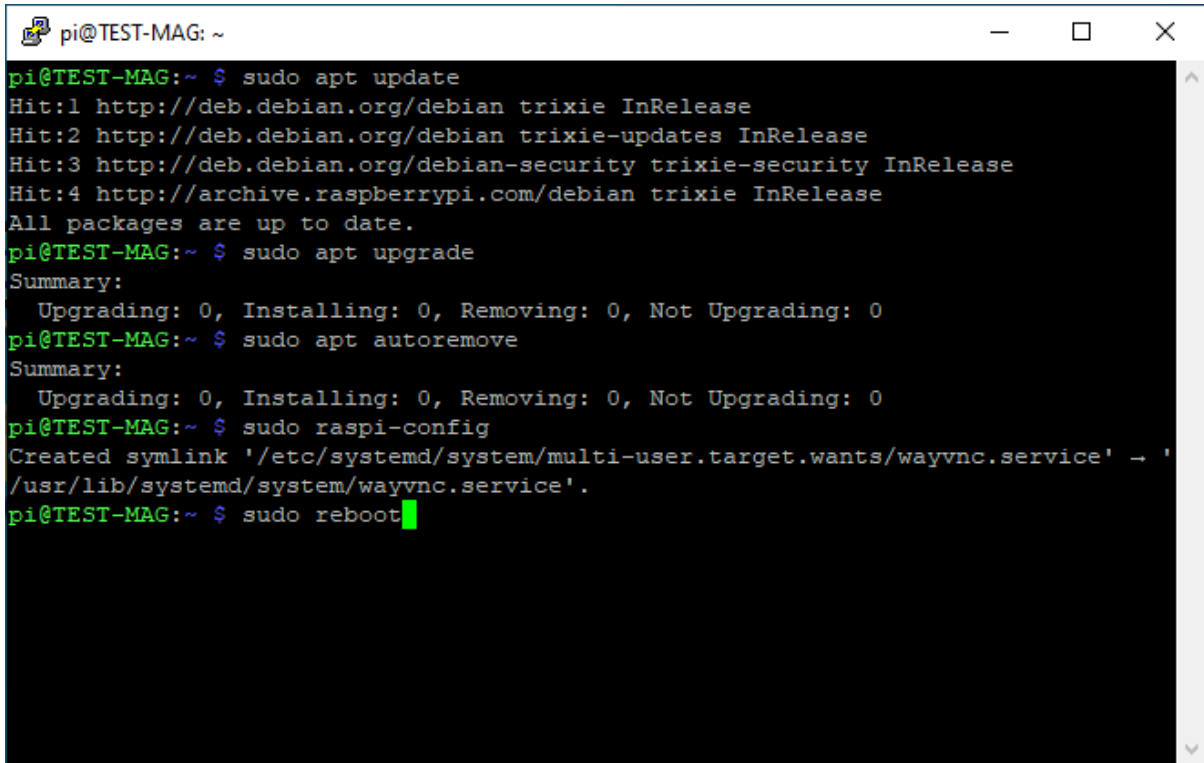


21. Finish with the Configuration Tool by selecting **Finish** and press **enter**.



22. Reboot the RPi, type the command below and press **enter**. This will close the PuTTY connection. You can close the Putty window.

sudo reboot

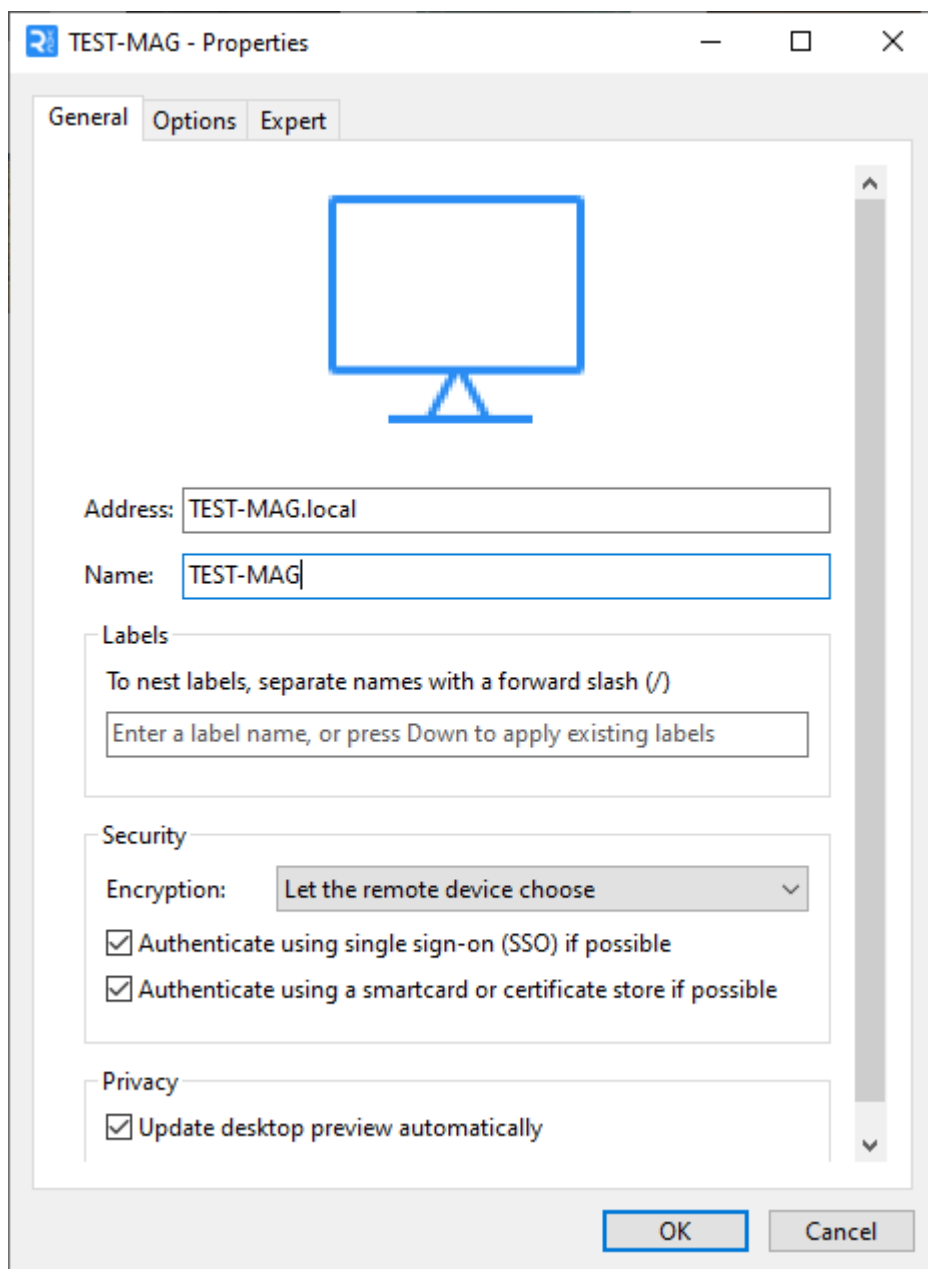


```
pi@TEST-MAG: ~  
pi@TEST-MAG:~ $ sudo apt update  
Hit:1 http://deb.debian.org/debian trixie InRelease  
Hit:2 http://deb.debian.org/debian trixie-updates InRelease  
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease  
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease  
All packages are up to date.  
pi@TEST-MAG:~ $ sudo apt upgrade  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
pi@TEST-MAG:~ $ sudo apt autoremove  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
pi@TEST-MAG:~ $ sudo raspi-config  
Created symlink '/etc/systemd/system/multi-user.target.wants/wayvnc.service' →  
'/usr/lib/systemd/system/wayvnc.service'.  
pi@TEST-MAG:~ $ sudo reboot
```

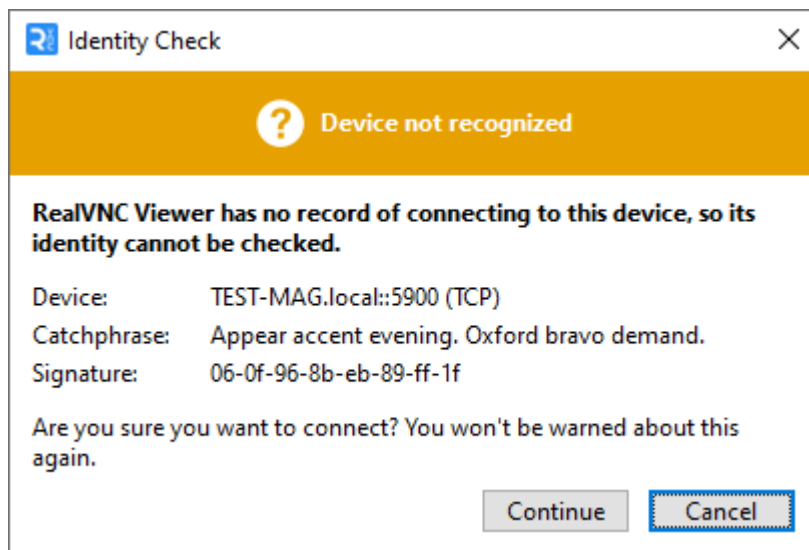
Headless access to RPi via VNC

Using a VNC program like RealVNC (<https://www.realvnc.com/en/>), connect to your RPi using your desktop PC...

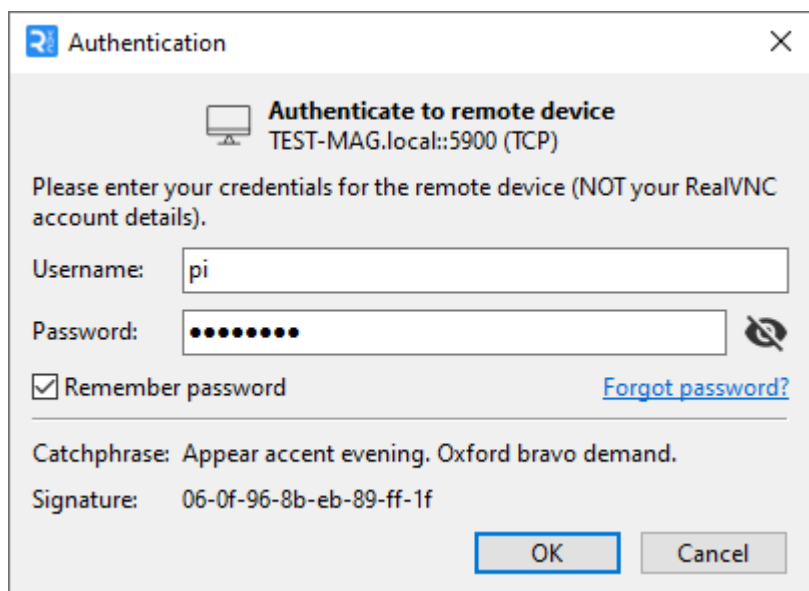
1. Create a **New connections** (CTRL-N) and enter the hostname into the **VNC Server:** box, add friendly name in the **Name** box and select **OK**.



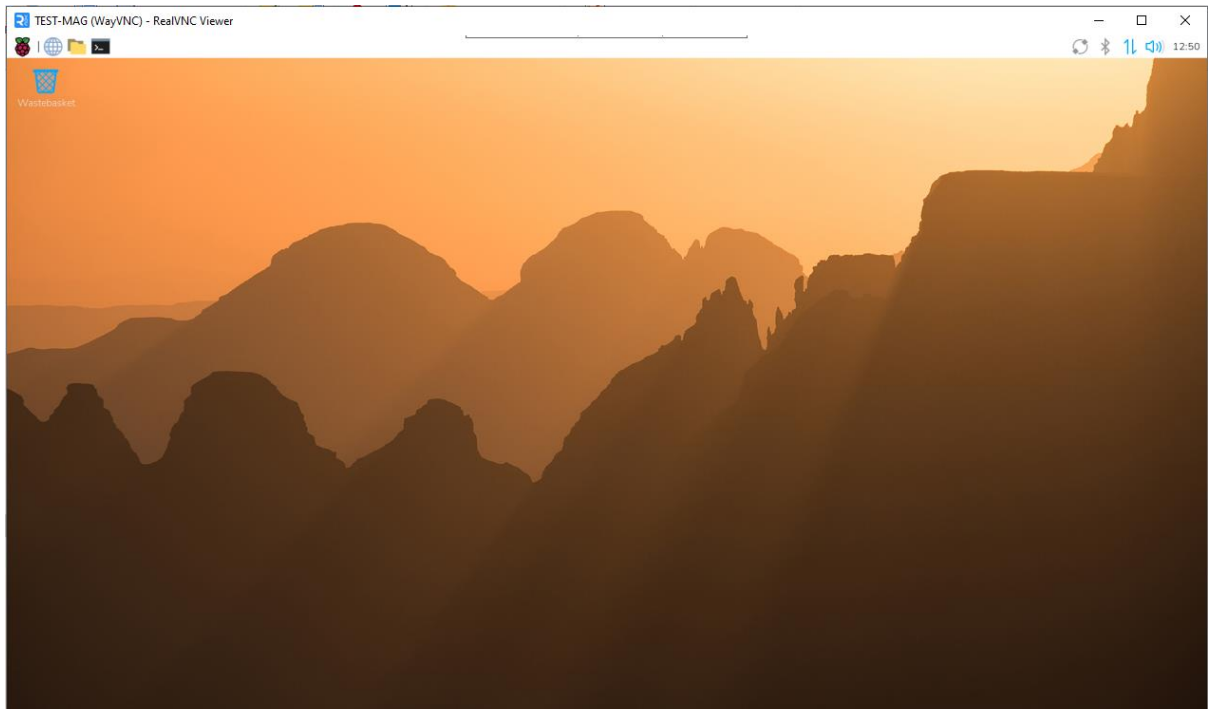
2. Open this VNC connection to your RPi; you will get a **VNC server not recognised** window from RealVNC. Select **Continue**



3. You will now get an **Authentication** window from RealVNC, enter your **Username** and **Password** details and select **Remember password** to make it easier to login in future.



4. You are now remotely accessing your RPi from your desktop PC...



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Contact us

Please send an e-mail to PicoMagnetometer@ukraa.com

